# Particles from Native Overlap Transport: Common Currents, Defect Modes and a Pixel-Dependent Mass Operator

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**Abstract**

Observer Patch Holography proposes that particles are reproducible features of the records of one specified observer, repair and memory process, and that stable massive matter consists of localized defects that repair cannot erase. We give the particle content of that process and compare it with measurement. Complete-response premises fix the Standard Model gauge type, and a declared lift gives one chiral generation. On a specified interacting cube, exact enclosures give 228 charged lines visible to single-fermion probes and 138 composite lines invisible to them, and a compiled instrument detects the composite response. The same localized modes carry all twelve native currents and a pixel-dependent charged-lepton matrix realized by complete memory dynamics. A compact opposite-winding texture has finite energy and four isolated modes, and its quartic extension has exact interacting poles. The specified mass branch gives twelve comparison values. Charged leptons lie within 71 parts per million under a pole premise and a target-informed law; the W, Z, Higgs and top lie within one percent; five lighter quarks miss by 23 to 209 percent. All twelve lie outside their printed intervals. We prove the exclusions this comparison supports and list what a source-selected spectrum must supply.

## 1 The particle question

Observer Patch Holography (OPH) proposes a self-referential complete history: observers infer the specification of their world from shared records and realize it through operations within that same history. The executable version specifies a carrier Hilbert space, accessible algebras, repair instruments, a scheduler, memory, feedback and preparation services [2, 6]. These are permitted design choices. Operational agreement concerns accepted shared data; state selection and endogenous implementation of response transformations are additional stated principles, recorded with the exact theorem premises in the common registry [28].

The source specification is a tuple

$$\mathcal{S} = (\mathcal{H}, \mathcal{A}, \{\Phi_{e,r}\}, \sigma, \mathcal{M}, \mathcal{F}, \mathcal{P}),$$

where $\Phi_{e,r}$ is the operation on seam $e$ with retained outcome $r$, $\sigma$ selects operations, $\mathcal{M}$ is memory, $\mathcal{F}$ is feedback, and $\mathcal{P}$ is the preparation domain. A particle assertion concerns the joint and

conditional laws of actual preparations, interventions and detector outputs of $\mathcal{S}$. Two equal spectra do not identify the states, currents, clocks or instruments that produced them, and Section 7 proves this in several exact forms.

**Hypothesis 1.1** (Matter as localized defects)**.** *Stable massive matter admits a localized excitation of a continuing source reference, with an invariant preserved by the admitted dynamics. Its creation, transport, interactions and detection are operations of the same source. Its total excess energy and propagation use one energy and clock dictionary. Unstable species may be metastable or internal excitations; massless radiation need not carry a defect invariant.*

The hypothesis does not ask for a distinct conserved topological charge on every mass eigenstate. Flavor mixing and decay can connect internal states of a defect family, and a protected label cannot change at an admitted vertex without another field or record carrying the compensating charge. Section 2.4 gives the exact constraint.

## 1.1 Main results

Table 1 states what this paper establishes. Three kinds of statement occur and we keep them apart. Some results follow from the unchanged source with every outcome retained. Some are exact on a chosen design, whose inputs are stated. Some compare a specified branch with measurement, and these are conditional postdictions.

Table 1: Main results and the inputs each one consumes.

| **Result** | **Inputs** |
|---|---|
| Internal charges of one chiral generation, three Higgs lines and the conditional quadratic mass table (Section 2) | Complete response, inner rotations, spin lift, three family copies, canonical kinetic forms |
| Complete transfer spectrum of the native carrier, every compact core among 4096 port sets, a positive-excess phase excitation and a positive distance from every quadratic (Section 3.1) | Uniform seam selection and the fair coin; no fitted coefficient |
| Certified charged spectra of the interacting cube, 228 linear lines and 138 composite lines, and a compiled composite detector (Section 3.2) | Couplings $c = 1/8$, $U = 1$; imported Lieb uniqueness; pulse services |
| Complete classification of conserved quadratics (Theorem 3.1) | Open connected box, positive hopping and repulsion |
| A complete memory realization of the charged coefficients and certified pixel feedback roots (Section 4) | Target-informed mass law; declared probes, gain and scale design |
| One localized algebra for currents and masses, a compact finite-energy pair, a gauge-covariant paired band, exact interacting poles and a controlled work operator (Section 5) | Chosen textures, coherent pairing service and quartic coupling |
| Twelve comparison values against the 2026 Particle Data Group tables (Section 6) | Target-informed branch, external unit, supersymmetric running proxy, imposed criticality |
| Exact exclusions of mass attachments and record-based selection rules (Section 7) | Each states its own domain |

No result in this paper derives a measured particle mass, and the construction is a precise conditional particle model. Section 8 lists what a source-selected spectrum must supply.

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# 2 Internal quantum numbers

## 2.1 Complete response and the compact Lie algebra

The carrier is the oriented icosahedral boundary, with twelve ports, thirty seams and twenty faces. Its real port module under the proper rotation group $A_5$ decomposes as

$$V_{12} = \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}, \qquad \dim V_{12}^{A_5} = 1.$$

Assume a faithful real-linear response map $D : V_{12} \to \mathfrak{u}(\mathcal{K})$ whose image is the complete connected response algebra and is closed under commutators, with proper carrier rotations acting through the same connected inner response group up to operationally silent centralizer factors. The gauge paper [5] proves that the compact classification then gives

$$\mathfrak{g} \simeq \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3), \tag{1}$$

because inner rotations fix a centre of dimension at most one and a centre-free $\mathfrak{su}(2)^4$ cannot have a one-dimensional fixed space, which leaves the semisimple split $11 = 3 + 8$. It also constructs an explicit current law on the protected native carrier realizing $\mathfrak{u}(3) \oplus \mathfrak{so}(3)$. Twelve ports alone do not imply (1), and carrier rechartings are distinct from physical spatial rotations of a relativistic scattering theory. This paper consumes the result and its twelve current matrices.

## 2.2 The matter module and its family factor

A declared spin lift to five internal coordinates and the nonvacuum even exterior module give

$$M_{15} = \Lambda^2\mathbb{C}^5 \oplus \Lambda^4\mathbb{C}^5, \qquad \dim M_{15} = 10 + 5 = 15,$$

the familiar $\mathbf{10} \oplus \overline{\mathbf{5}}$ branching [9]. The source-current lift preserves all brackets and has an exact intertwiner with this module. Its simultaneous primitive-charge and color/weak Casimir readouts give the following all-left convention.

| Label | Color | Weak | $6Y$ | Components |
|---|---|---|---|---|
| $Q$ | $\mathbf{3}$ | $\mathbf{2}$ | 1 | 6 |
| $u^c$ | $\overline{\mathbf{3}}$ | $\mathbf{1}$ | $-4$ | 3 |
| $d^c$ | $\overline{\mathbf{3}}$ | $\mathbf{1}$ | 2 | 3 |
| $L$ | $\mathbf{1}$ | $\mathbf{2}$ | $-3$ | 2 |
| $e^c$ | $\mathbf{1}$ | $\mathbf{1}$ | 6 | 1 |

The labels abbreviate computed finite quantum numbers; their identification with laboratory Weyl fields requires the physical current, spin and kinetic maps. The gauge paper [5] gives the anomaly scan, which selects this module among exterior masks up to conjugation, the $\mathbb{Z}_6$ global form and the conditional family theorem. For the particle construction we explicitly choose $M_{15}\otimes\mathbb{C}^3$, with gauge currents acting trivially on $\mathbb{C}^3$. A two-dimensional radial kernel or a winding number three is no proof of three generations, and equivalent triplet representations on spatial, color and family factors do not identify those factors.

Exchange records have a separate finite route. On the native ordered-cover transport, let $p_h(x)$ be the permutation parity accumulated by branch $h$. A coherent ordering bit gives

$$K_h = \frac{1}{\sqrt{2}} \sum_{x\in D_h} |f_h(x)\rangle\langle x| \otimes X^{p_h(x)}, \qquad Q = I \otimes X, \qquad [K_h, Q] = 0.$$

An odd exchange loop acts as $X$ on the bit, and the corresponding colored fermionic lift acts as $-$SWAP. The sectors survive complete record-adaptive native schedules in this specified memory design; dephasing or resetting the bit changes that conclusion. These are exchange-history sectors, distinct from the spatially localized texture charge of Section 5.

### 2.3 The conditional field action

Choose one complex weak doublet $H$ of hypercharge $1/2$, canonical positive kinetic forms, physical Weyl statistics and a Lorentz-covariant local field interpretation in $3+1$ dimensions. Restrict the polynomial action to operator dimension at most four, modulo integration by parts. The invariant grammar then gives the gauge kinetic terms, scalar kinetic term and potential, fermion kinetic terms and exactly three charged Yukawa lines. Independent gauge couplings, topological theta terms and an additive vacuum-energy constant are also free. In the four-component convention the charged Yukawa part is

$$\mathcal{L}_Y = -\overline{Q}_L \widetilde{H} Y_u u_R - \overline{Q}_L H Y_d d_R - \overline{L}_L H Y_e e_R + \text{h.c.}$$

Each $Y_a$ is a complex $3\times 3$ matrix. The native invariant-line computation solves all twelve current equations for both Higgs components and gives one line for each allowed origin $(6Y_1, 6Y_2) = (-4,1),(1,2),(-3,6)$. Tensoring these lines with the gauge-trivial family factor leaves exactly the three matrix coefficients. Canonical kinetic normalization must precede any reading of their singular values as action masses [5, 26, 25].

For the explicitly chosen broken potential

$$V(H) = \lambda_H (H^\dagger H - v^2/2)^2, \qquad \lambda_H > 0,$$

the vacuum shell has $|H| = v/\sqrt{2}$. Taking $H = (0, v/\sqrt{2})$ gives the following conditional quadratic table.

| Field directions | Action mass or coefficient |
|---|---|
| Three charged leptons | $m_{e,i} = v\, s_i(Y_e)/\sqrt{2}$ |
| Three up and three down flavors | $m_{a,i} = v\, s_i(Y_a)/\sqrt{2}$, each in three colors |
| Three minimal left neutrino fields | Zero in the dimension-four charged grammar |
| Charged weak vectors | $m_W = g_2 v/2$ |
| Neutral massive weak vector | $m_Z = v\sqrt{g_2^2 + g_Y^2}/2$ |
| Electromagnetic direction | Zero quadratic mass |
| Eight color directions | Zero quadratic mass; no free physical gluon claim |
| One radial scalar | $m_h = \sqrt{2\lambda_H}\, v$ |

The neutral vector matrix follows from the coefficient of $\frac{1}{2}A^T M^2 A$ in $|DH|^2$; this factor of two matters. The three zero scalar Hessian directions are gauge-orbit directions. Neither the sign of the broken potential nor its quantum vacuum is selected by the finite representation.

For a supplied Weyl principal symbol and the antisymmetric pairing $\epsilon_{\text{spin}} \otimes \overline{K}$, squaring the complete Nambu symbol cancels its momentum–mass cross terms, so

$$E_j(\mathbf{p})^2 = c^2|\mathbf{p}|^2 + s_j(K)^2.$$

This is a theorem of that supplied relativistic action. The spatial source of Section 5 has additive bands instead, and the formula does not transfer to it by matching $K$ alone.

### 2.4 Flavor, neutrinos and conserved defect labels

The canonical family-frame freedom is $\mathrm{U}(3)^5$, and it classifies every same-current, canonical-kinetic family embedding. It preserves the charged singular spectra and basis-invariant mixing data. In particular $V_{\text{CKM}} = U_{uL}^\dagger U_{dL}$ is determined by the same left-handed weak current; relabeling families cannot repair a mass hierarchy or generate physical CP violation. A real symmetric coefficient class has $\mathrm{tr}[Y_u Y_u^\dagger, Y_d Y_d^\dagger]^3 = 0$ [24].

The native current also supplies one $LH$ invariant line, and its symmetric family lift enters the complete action with six free complex Weinberg coefficients. A supplied dimension-five operator $(L \cdot H)^T C_5 (L \cdot H)/(2\Lambda) + \text{h.c.}$ gives $M_\nu = v^2 C_5/(2\Lambda)$ and Takagi singular masses. A calibrated instance has one massless neutrino and two nonzero coefficients. The charged-lepton construction supplies neither $C_5$ nor its scale.

Conserved additive defect labels must respect the same vertices. If the charged-current mixing graph is connected, the relation

$$q(W^-) - q(e_\alpha) + q(\nu_i) = 0$$

for every nonzero entry forces one common neutrino label $N$ and one common charged-lepton label. A massive Majorana field adds $2N = 0$, which in an integer winding group implies $N = 0$. If a massless neutrino is excluded from defect assignments and has label zero, connected mixing forces the other neutrino labels to zero as well. Distinct protected winding numbers therefore cannot simply be assigned to the three mixed mass eigenstates. Internal excitations, nonadditive invariants and charged compensating vertices require their own source construction.

# 3 The spectrum of the native source

Before any mass law is chosen, the source has spectra of its own. This section collects them. No coefficient here is fitted to a particle, every energy is in the source's own dimensionless unit, and each result retains the complete state space on which it is stated.

The native source acts on twelve nonnegative integer port loads of total two, which form 78 configurations with an invariant shell of 66 binary ones. For an oriented seam $(a, b)$ with contrast $d = x_a - x_b$ and a fair placement coin $c$, repair moves $j = \frac{1}{2}[d - (2c-1)(|d| \bmod 2)]$ units from $a$ to $b$. Instrument C records $(c, j)$ and instrument E also records the seam-total parity; they share one classical update and differ in the coherence they retain [6].

## 3.1 Transfer modes, compact cores and a phase excitation

Keep all sixty-six binary total-two configurations. Uniform independent seam selection and fair placement give the complete attempt transfer $P = C/60$, with $2I/3 \le P \le I$, from all 3960 endpoint branches. Exact characteristic polynomial factorization, independently checked by sixty-six Newton traces, finds sixteen distinct eigenvalues and a unique constant ground vector. The largest nontrivial eigenvalue is $(55 + \sqrt{5})/60$.

A compact amplitude core means that both loads lie in a selected port set. The largest confined invariant subspace is the kernel of the outgoing block and all its products with powers of the internal block. Exhausting all 4096 subsets, with exact rank certificates, gives no nonzero such subspace on nine or fewer ports. Six ten-port cores support three modes, with transfer eigenvalues 4/5, 5/6, 13/15 and integrated normalized native autocorrelations 5, 6, 15/2 attempts. Admitted changes of seam weights destroy their confinement, and the full recorded instruments do not conserve their transfer sector. A common invariant amplitude subspace under every native seam uses all twelve ports.

The faithful reconstructed reference $\rho = P/\operatorname{Tr} P$, with $\operatorname{Tr} P = 56$, has a different localized excitation. Let $D = (-1)^{n_0}$ and $\rho' = D\rho D$. It has the same entropy, configuration probabilities and full eleven-port exterior as $\rho$. For the source-derived $H = -\log P$,

$$\frac{1}{1008} \le \operatorname{Tr}(H(\rho' - \rho)) \le \frac{1}{672}.$$

The bounds follow from $\|[P, D]\|_{\mathrm{HS}}^2 = 1/9$ and the positive spectral bounds on $P$. The native C and E unread channels contract the state contrast by 5/6 and 7/10 per uniform attempt, and complete tape-plus-exterior laws agree when compared with the equally evolved reference. The excitation is the reconstructed functional $A \mapsto \ell(DAD)$ and a decaying finite excitation; no native phase gate or coherent preparation is inferred from it.

Interaction is detectable without a fitted spectrum. Every quadratic operator in this fixed occupation dictionary has diagonal $K_{\{i,j\},\{i,j\}} = c + \epsilon_i + \epsilon_j$, regardless of its hopping terms, so its four-pair contrast on $01, 23, 02, 13$ vanishes. For $I - P$ the same contrast is $-1/15$, which puts it at norm distance at least $1/60$ from every such quadratic. A positive logarithm expansion with an explicit tail bounds the distance for $-\log P$ below by 0.01984562. The test retains every native state and permits arbitrary quadratic hoppings, on-site potentials and an energy offset. Arbitrary re-encodings lie outside it.

### 3.2 The charged spectrum of the interacting cube

The open eight-site Weyl/contact cube is the smallest interacting source with a complete charged spectrum. In its exact hidden-spin frame the source is

$$K = \mathrm{d}\Gamma(A \otimes I_2)/16 + \sum_x n_{x\uparrow} n_{x\downarrow} - N/2, \qquad A = A^T, \quad A^2 = 3I,$$

at $c = 1/8$, $U = 1$. All source matrices come from the full 65,536-state integer Hamiltonian. The reference is the unique half-filled ground, with hidden spin zero, eta zero and energy in $[-4.12405873337830, -4.12405873337828]$. Its uniqueness uses the theorem of Lieb on the half-filled Hubbard model [10] together with a full-number-sector bound. Hidden spin and eta are internal algebras, and neither carries a Lorentz-spin assignment.

**Lines visible to single-fermion probes.** Every local one-spin addition or removal measure has exactly 228 distinct positive energies, each with rigorously positive weight. The signed retarded measure has 456 atoms, each charge branch has total weight 1/2, and every eight-site residue matrix has visible rank four per hidden spin. The smallest weight is near $5.63 \times 10^{-21}$, and no threshold discards a line.

**Lines invisible to them.** Combine a local fermion removal with a neighboring hidden-spin flip,

$$C_x = \Big(\sum_{y \sim x} S_y^+\Big) c_{x\downarrow}, \qquad S_y^+ = c_{y\uparrow}^\dagger c_{y\downarrow}, \qquad \|C_x\| \le 3.$$

On the singlet reference the output has total hidden spin 3/2 and eta 1/2. The linear probes have hidden spin 1/2 and $K$ commutes with hidden spin, so the complete dynamical cyclic spaces of the two responses are orthogonal. Exact integer residuals certify all 138 spectral lines in the composite probe's reducing block, each with strictly positive weight, and direct calculations of $C^\dagger C$ and $C^\dagger[K, C]$ on the unreduced source independently enclose the total weight and energy moment.

Table 2: Certified charged lines of the centered cube, in source-clock energy units. The first six of each spectrum are shown; the complete tables retain every line and weight.

| Line | Linear probe | Relative to first | Composite probe | Relative to first |
|---|---|---|---|---|
| 0 | 0.361682711806 | 1.000000000 | 0.402468515770 | 1.000000000 |
| 1 | 0.402293044056 | 1.112281652 | 0.422089756328 | 1.048752237 |
| 2 | 0.415174263333 | 1.147896346 | 0.438990742922 | 1.090745551 |
| 3 | 0.438196496809 | 1.211549467 | 0.479938217500 | 1.192486365 |
| 4 | 0.462192207579 | 1.277894111 | 0.485785522474 | 1.207014967 |
| 5 | 0.474429777956 | 1.311729210 | 0.494035650369 | 1.227513783 |
| Largest | 3.886434754951 | | 2.974756508168 | |
| Conditional mean | 0.49955163275 | | 0.52537938729 | |

Each composite residue matrix is twice its local weight times one of the source's two kinetic-band projectors, of exact rank four, with 69 lines on each band. Spin and eta ladders give 2,208 visible composite states per charge sector. Their orthogonal union with the linear probes covers 4,032 of the 11,440 states of that number sector, and the block therefore does not exhaust every charged composite. Independent source equations and all 192 spatial energy moments retain the

contact contribution and the kinetic/contact interference. The finite spatial projectors supply no continuous momentum dispersion.

The composite response can be read. A compiled pulse instrument implements the success Kraus operator $C_0/3$ in 1,008 source pulses on the authenticated hidden-frame ground preparation, retaining all 7,776 raw selector records, native aborts and coherent field-reference amplitudes. Its success probability is approximately 0.08261623248916, one ninth of the certified composite spectral weight. It detects the total response; no clock resolving the 138 energies is executed.

These lines give one clean exclusion. The visible energy spread is below eleven, so no three distinct lines with one positive unit reproduce the charged-lepton mass ratios. That excludes this finite gap dictionary, and it leaves other sources, refinements and unseen composites untouched.

### 3.3 Complete conserved quadratics

The labels of these spectra come from a complete classification.

**Theorem 3.1** (Complete quadratic conservation)**.** *On a finite connected open rectangular box with $V$ sites and two fermion modes per site, let*

$$H_\mu = \sum_{\langle x,y=x+e_j\rangle} c_x^\dagger (\tfrac{ic}{2}\sigma_j) c_y + \text{h.c.} + U\sum_x n_{x\uparrow}n_{x\downarrow} - \mu N, \qquad c, U > 0.$$

*Modulo scalars, the real space of self-adjoint even conserved operators of degree at most two is spanned by $N - V$ and $S_1, S_2, S_3$ when $\mu \neq U/2$, and by these together with $\eta_x, \eta_y$ when $\mu = U/2$, where*

$$S_a = \tfrac{1}{2}\sum_x (-1)^{\sum_{j\neq a} x_j} c_x^\dagger \sigma_a c_x, \qquad \eta_+ = \sum_x (-1)^{x_1+x_2+x_3} c_{x\uparrow}^\dagger c_{x\downarrow}^\dagger.$$

*The traceless conserved Lie algebra is hidden $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$ for $H_0$ and hidden $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ for the centered generator $K = H_{U/2}$.*

*Proof.* Commutators with $H_\mu$ preserve the number change of each term, so the number-preserving and anomalous parts separate. The one-particle restriction gives $[h, q] = 0$ for a one-body matrix $q$, hence kinetic commutation on the whole Fock algebra and commutation with the doublon count. An input with one spectator and its doublon output isolate each intersite coefficient and force it to vanish. The kinetic condition $q_y = \sigma_j q_x \sigma_j$ then transports one Hermitian root matrix over the connected box, which gives the four stated coefficients. For a pair creator, comparing the vacuum-to-pair entry with its spectator entry gives $U\Delta_{pq} = 0$ on distinct sites. Onsite terms obey $(U - 2\mu)f_x = 0$, and at $\mu = U/2$ the kinetic condition forces the alternating profile, whose real and imaginary parts are $\eta_x$ and $\eta_y$ [11]. □

Exact full-CAR coefficient ranks and independent occupation-matrix checks verify the finite instances, and centering changes the conserved pairing algebra as well as the charged energies. Higher-degree currents, Lorentz stress and an interacting chiral continuum are outside the theorem. The radial defect source has its own classification: every conserved sum of onsite even operators is eta-pairing SU(2), and on the mass-64 source every center Pauli current has leakage norm above 1/64 outside the low quartet. The complete native color, weak and hypercharge action therefore cannot be inherited from that onsite-even algebra.

### 3.4 One-body massless and massive sectors

The same source has exact one-body sectors. On the periodic cube of eight sites per axis a complete spectral detector resolves all 1,024 one-particle modes, including sixteen zeros and six pairs of signed shells, with every pointer and abort outcome retained. A stationary one-particle state cannot occupy only six sites, and the infinite lattice has no square-summable stationary one-particle state.

With centered clock $H-(U/2)N$ and $c = v/a$, the continuum is the free massless field in all eight Weyl valleys, $E = v|\mathbf{p}|$, throughout $0 \le U$, $U = o(c^{2/7})$ [6]. Supplied quadratic mass coefficients give two Dirac copies at each rest energy $|m_0 \pm |\mathbf{m}||$, with continuum dispersion $E^2 = v^2|\mathbf{p}|^2 + M^2$ and a proved finite-spacing error. This uniform family has at most two distinct positive rest energies, so a direct assignment cannot reproduce three distinct charged-lepton masses. The ordered-spin continuum has nonzero positive-energy bands at every small momentum. Complete energy–symmetry readouts bound the original response near a specified dispersion, with aliases, clock leakage and all failures retained. Selection-field and zero-field histories have a same-input comparison bound; a finite prepared state is not identified with the infinite ordered vacuum. Relaxed inverse susceptibility determines an infinitesimal-response cost, not a particle mass. A uniform nonvanishing shrinking-tube bound, Lorentz covariance and surviving continuum interactions are unproved. Matching a light speed fixes length unit $= cT$; determining $T$ requires an independently identified same-generator clock.

## 4 The charged mass operator

### 4.1 Pixel capacity and the mass law

The pixel variable is the dimensionless capacity ratio $P = a_{\mathrm{cell}}/\ell_P^2$, distinct from a projector $P_j$ in the memory controller. The declared self-reference relation reads

$$P = \varphi + \sqrt{\pi}\,\alpha, \qquad \varphi = \frac{1+\sqrt{5}}{2}.$$

In the heat/running construction the inner response supplying $\alpha$ depends on $P$ through a unified scale, a heat-sum equation, low-energy couplings and charged masses, so the loop has the form

$$P \longrightarrow (\alpha_U, v, \text{field census}) \longrightarrow (M_f, J_f) \longrightarrow A_{\mathrm{response}} \longrightarrow \varphi + \frac{\sqrt{\pi}}{A_{\mathrm{response}}}.$$

The same masses and current normalization appear on both sides. A fixed point of a declared response proves consistency of that map, and its identification with the physical electromagnetic response is a further claim.

The controller is a native family memory. It stores one of twelve port addresses, each with a rank-one family projector $P_j$. A C event rotates the port label, a joint event then applies the rotation of E, and other records reset the address to port two. The fair-coin rates of the C and E events are $p = 1/12$ and $q = 1/6$, and all 128 raw record labels are retained. Declared current probes lower these rates to the values that follow.

The mass law used here is a chosen response architecture with icosahedral registers and determinant normalization. Its charged shape uses a three-phase Koide-type parametrization [7, 8]. The architecture has target-informed provenance: its numerical comparisons are conditional postdictions even when its execution reads no measured masses. The quark exponents, scalar quartic, neutrino operator and physical flavor mixing are not determined from the premises A1–A3 of the registry [28]. Independent validation requires consequences not used to choose it. The operator

realization below applies to the displayed $\beta = 4$ branch and to the normalized coefficients of the explicit $\beta = 7$ branch.

### 4.2 A pixel mass law in native memory dynamics

A chosen simulator specification can realize the conditional pixel-dependent charged mass law on the native gauge-family carrier. The construction uses the twelve-port reset controller and the complete native current instruments. Its design inputs are independent left and right probe durations, a rule enabling feedback on a successful C event, and a common feedback amplitude. The conditional mass-law architecture is target-informed; evaluating that architecture without a charged-mass input does not establish a prospective prediction.

The selected response architecture gives

$$\kappa = \frac{\alpha_U/50}{1+\alpha_U/31+\alpha_U^2/310}, \qquad \chi = \frac{\alpha_U^2/512}{1-\alpha_U/77}, \tag{2}$$

$$\zeta = \frac{(P-\varphi)^2/(21\pi)}{1-\alpha_U/27-\alpha_U^2/135}, \qquad r_j = 1+\sqrt{2}e^{-\chi}\cos\left(\frac{2}{9}+\zeta+\frac{2\pi j}{3}\right), \tag{3}$$

where $\varphi = (1+\sqrt{5})/2$ and $j = 0, 1, 2$. Its charged coefficients, ordered increasingly, are

$$t_i = \frac{m_i}{v} = \frac{e^{\kappa} r_i^2}{\left(2\,6^{14}\prod_{j=0}^{2} r_j^2\right)^{1/3}}, \qquad \frac{v}{E_*} = P^{-1/2}\exp\left(-\frac{2\pi}{4\alpha_U}\right). \tag{4}$$

The register dimensions and determinant exponent in these formulas are part of the selected architecture. The scale $E_*$ is a source coordinate; assigning it an energy in laboratory units requires a clock calibration.

Let $t_i > 0$ be these three charged coefficients and $\lambda_i = t_i/\sum_j t_j$. For the native projectors $P_2, P_3, P_{10}$, direct multiplication gives $\operatorname{tr}(P_iP_j) = 1/5$ for distinct indices and the spanning-vector Gram determinant $d = 2/5 - 2\sqrt{5}/25$. Consequently the matrix $Y = uP_2 + v_1P_3 + wP_{10}$ has elementary eigenvalue invariants

$$\operatorname{tr} Y = u + v_1 + w, \qquad e_2(Y) = \frac{4}{5}(uv_1 + uw + v_1w), \qquad \det Y = duv_1w. \tag{5}$$

Choose its three positive weights as the roots of

$$z^3 - z^2 + \frac{5}{4}e_2(\lambda)z - \frac{e_3(\lambda)}{d} = 0, \qquad w < u < v_1. \tag{6}$$

It follows that $\operatorname{spec} Y = \{\lambda_i\}$.

The complete C-enabled memory coefficient, with $a = p(1-q)$, $b = pq$ and $D = 1 - a^2 + ab$, is

$$Z_C = \frac{(1-a)P_2 + a(1-a)P_3 + abP_{10}}{D} - (1-p)P_2. \tag{7}$$

It has trace $p$; disabled records have not been removed or normalized away. Put $A = v_1(v_1+w)$ and $B = v_1(1-v_1+w) + (v_1+w)w$. The inverse equations

$$-Ap^2 + Bp - w = 0, \qquad q = 1 - \frac{v_1}{1-p(v_1+w)} \tag{8}$$

give the exact identity $Z_C = pY$. This follows by substituting the second equation and reducing each of the three coefficient differences by the quadratic in the first equation.

At each of the supplied forward and measured-endpoint pixel coordinates, rational Taylor enclosures and polynomial endpoint signs give exactly three roots of (6), and exactly two solutions of (8) satisfying $0 < p < 1/12$, $0 < q < 1/6$. For the forward coordinate the first solution is approximately $p = 0.01915526454$, $q = 0.05474497127$. The native charged-lepton support Casimirs are $9/8$ and $3$, so the complete probes implement these rates with the declared durations

$$\theta_L = \frac{\arcsin\sqrt{12p}}{\sqrt{9/8}}, \qquad \theta_R = \frac{\arcsin\sqrt{6q}}{\sqrt{3}}. \tag{9}$$

Both probes retain all thirteen outcomes. Their product retains all $169 \times 128$ current and repair labels per old memory input. The covariant binary grouping retains 512 labels per input and all twelve old inputs.

Choose $g = \sum_i t_i/p$. Then $gZ_C$ has exactly the three coefficients $t_i$. Its diagonalizing unitary acts on the gauge-trivial family factor, commutes with every native gauge current, and preserves the canonical kinetic form. Thus this is an operator realization of the conditional mass law, up to the allowed family frame. It does not assert an additional alternating-group intertwiner. The source-coordinate enclosures also give $\sum_i t_i < 1$ by bounding the cube of that sum.

The coefficient is the generator of a controlled limit of the full correlated memory. Apply $\exp(-ihgX \otimes P_{\text{new}})$ on enabled records and identity on disabled records. Let $\Psi_h$ be this complete classical/quantum channel. Its classical transition satisfies $T = (1-p)R + pS$, where $R$ resets to port two and $S$ is stochastic. For every normalized initial classical/quantum state $z_0$, including an arbitrary untouched finite reference, the reduced state after $N$ steps of size $h = \tau/N$ obeys

$$\left\| \sum_{\text{memory}} \Psi_{\tau/N}^N(z_0) - \mathcal{U}_\tau\left( \sum_{\text{memory}} z_0 \right) \right\|_1 \le \frac{4g\tau}{N(1-p)} + \left( \frac{8}{1-p} + 4 \right) \frac{g^2\tau^2}{N}, \tag{10}$$

where $\mathcal{U}_\tau$ is generated by $gX \otimes Z_C$. For a stationary product preparation the first term vanishes.

To prove the bound, let $\Pi z = \pi \sum z$ and $e_k = (I - \Pi) z_k$. The sum of trace norms on memory blocks gives $\|Te\|_1 \le p\|e\|_1$ on zero-sum blocks and $\|\Psi_h - T\| \le 2gh$. Hence $\|e_k\|_1 \le 2p^k + 4gh/(1-p)$. The reduced stationary step and the mean unitary step have the same first derivative; their two Taylor remainders sum to at most $4g^2h^2$. The reduced error due to $e_k$ is at most $2gh\|e_k\|_1$. Summing these errors proves (10).

This limit is in dimensionless simulator time. A finite unread channel need not equal its limiting unitary. The feedback pulse service is an explicit design choice; compilation into a smaller native gate alphabet is a separate requirement. The forward pixel coordinate has no complete physical endpoint proof, and the measured-endpoint coordinate inherits measured-alpha calibration. No physical pole-mass identification, renormalization map, operational energy scale, or complete quark and neutrino spectrum follows from this finite realization.

### 4.3 Memory correlations beyond the mean unitary

The native stationary memory has three distinct projector rays, because $P_9 = P_{10}$. Summing their complete temporal correlations gives a controlled correction to the mean evolution. Let $T$ be the reduced column-stochastic transition, $E$ its enabled-pulse part, $\pi$ its stationary column, $\Pi = \pi \mathbf{1}^T$ and $w = E\pi$. The reset identity contracts zero-sum memory vectors by $p < 1$, so the group inverse $\mathcal{R} = (I - T + \Pi)^{-1} - \Pi$ sums every lag. The one-sided covariance and its positive symmetric part are

$$K = \tfrac{1}{2}(\operatorname{diag} w - ww^T) + E\mathcal{R} \operatorname{diag} w, \qquad D = K + K^T \succeq 0,$$

where positivity follows from the limiting covariance of accumulated pulse counts. For $H_i = gX \otimes P_i$ and $L_i = -i[H_i, \cdot]$, the reduced generator is $G_0 + hG_1$, with $G_0 = \sum_i w_i L_i$, $G_1 = \sum_{ij} K_{ij} L_i L_j$. Its antisymmetric coefficient gives the coherent ordering potential $H_{\mathrm{ord}} = -(i/2) \sum_{ij} K_{ij}[H_i, H_j]$; the symmetric part gives the dissipator $-\frac{1}{2} \sum_{ij} D_{ij}[H_i, [H_j, \cdot]]$. Both belong to the same correction, and $H_{\mathrm{ord}} = I_2 \otimes A$ is a family potential with a different auxiliary grading from a scalar mass.

For stationary product preparation, any untouched reference, $h = t/N$ and fixed $t$, the corrected completely positive semigroup has an explicit rational $O(h^2)$ finite-time diamond-norm bound on the unread channel. Complete record tapes and correlated initial memory need their own terms.

### 4.4 Complete records, capacity and recovery

The native twelve-address family memory admits an exact smaller encoding. Every raw branch preserves the recurrent set $S = \{2, 3, 9, 10\}$. Let $E_S$ embed four memory addresses into these source addresses. For all 128 raw labels, all twelve original old-memory labels and every real pulse size, the corresponding reduced branch $B_y$ obeys

$$K_y(E_S \otimes I) = (E_S \otimes I)B_y.$$

Multiplication proves equality of all joint records and terminal work states for any initially encoded state, including entanglement with an arbitrary reference. Reset to address 2 and stationary memory qualify. Address 0 admits one fully retained startup event, landing at 2 or 9, followed by exact recurrent encoding. These maps encode one complete law, and physical independence of capacity would need more.

The native matching code obeys the ideal recovery identity $R_y K_y E = \sqrt{w_y} E$, with $\sum_y w_y = 1$, for every retained branch, on both matching codes, both repair laws and all thirty seams. Specified logical evolution between corrected repairs is therefore preserved even conditional on the records, and faithful recovery cannot shift its poles.

Changing feedback duty changes the evolution. The native always-duty and C-duty generators have ratios approximately $1 : 195.117 : 3579.826$ and $1 : 53.588 : 271.403$. Their normalized quadratic traces differ exactly, so neither a family relabeling nor a common clock rescales one into the other. Dividing by event probability changes only the common scale. The specified nine-transition preparation and two-clock inverse have stationary-kernel error at most $1/2579890176$, which controls a first-order kernel approximation.

### 4.5 Flavor from current-gated memory

Reading a specified memory's angle, reset rule, duty policy and feedback law determines its stationary coefficients, and these are source-derived for that specification. The declared native current gates the complete repair record differently in the up, down and charged-lepton sectors, and the static mean-action service has certified full-action spectra and nontrivial real quark mixing. For $\theta = \pi/5$, duty "always" and unit Higgs norm, the dimensionless coefficients are as follows.

| Channel | Three ordered coefficients | | |
|---|---|---|---|
| Up | 0.00015764215 | 0.04279127122 | 0.95705108663 |
| Down | 0.00014014156 | 0.04369077654 | 0.95616908190 |
| Charged lepton | 0.00003179436 | 0.02148604675 | 0.97848215888 |

Every coefficient in the retained real-projector class has $\mathrm{Tr}[H_u, H_d]^3 = 0$, and complex family basis changes cannot create charged-current CP violation. All four default duty policies fail direct

identification of the charged-lepton ratios: under a common positive normalization and relative corrections bounded by $\delta$, the unconditional and C-duty cases require $\delta \geq 0.79167$ and $\delta \geq 0.59931$, and E and CE duties have an exact zero coefficient. Other controllers and source angles lie outside this comparison.

### 4.6 Pixel feedback with one charged-fermion mass prescription

The pixel feedback must use the same mass coefficients as the declared field and memory operators. Let $u = \alpha_U$ and retain

$$M_U/E_* = e^{-2\pi}P^{1/6}, \qquad v/E_* = P^{-1/2}e^{-2\pi/(\beta u)}, \qquad \alpha_i^{-1} = u^{-1} + \frac{b_i}{2\pi}\log(M_U/m_Z).$$

Here $E_*$ is a declared energy unit. The same conditional three-lepton coefficients are inserted into the charged-fermion integral, together with all six declared quark coefficients, including the finite top contribution. With the prescribed factor $1 - 3\alpha_3/\pi$ on the quark terms, the resulting inverse response $A(P)$ defines $P = \phi + \sqrt{\pi}/A(P)$. This kernel supplies no charged-vector or charged-scalar terms, confining hadronic spectral measure, threshold matching, or derivation of the native current's physical Thomson limit.

There is an exact obstruction to substituting the minimal Standard Model census while retaining $\beta = 4$. Assume $P \geq 1$, $M_U \geq m_Z$, positive couplings and threshold-free one-loop coefficients $b = (41/10, -19/6, -7)$. Set $x = \alpha_2^{-1}$ and $y = \alpha_3^{-1}$. Eliminating the unified coupling and scale ratio gives $0 < y \leq x$ and

$$\frac{\pi(18x + 5y)}{46} = 2\pi - \frac{2}{3}\log P + \frac{1}{2}\log\left[\pi\left(\frac{1}{x} + \frac{69}{333x - 218y}\right)\right].$$

The logarithm's argument is at most $8\pi/(5x)$. The function $9\pi x/23 - 2\pi - \frac{1}{2}\log[8\pi/(5x)]$ is increasing and exceeds one at $x = 6$, so $x < 6$ and both nonabelian couplings exceed $1/6$. Their heat-sum denominators are at least one; direct outward summation with infinite-tail bounds places their combined numerators below $1/80$. The required pixel entropy is $P/4 \geq 1/4$, a contradiction. This excludes the stated scale and census combination, independently of the readout kernel.

Two other declared maps have certified local fixed points. The supersymmetric running proxy with $\beta = 4$ gives $P \simeq 1.630971514159316$ and $A \simeq 137.000994766342$. The minimal census with the explicit scale choice $\beta = |b_3| = 7$ gives $P \simeq 1.628533369611833$ and $A \simeq 168.815082928453$. The first requires extra supersymmetric fields absent from the native quadratic carrier. The second supplies a scale choice, without identifying the Higgs scale physically with color transmutation. For each map, verified implicit-root signs, infinite heat tails and interval automatic differentiation give a strict self-map and $|g'| < 0.00734$ on the reported coupling interval. Uniqueness is local to that interval.

Rational interval inversion over both complete root enclosures admits both native memory solutions, with $0 < p < 1/12$ and $0 < q < 1/6$. The gain $g = \sum_j (m_j/v)/p$ preserves the unconditional duty probability, so the same full-record memory coefficient has the required three singular energies. Independent simultaneous-root and quadrature checks accompany the proof. These are conditional consistent feedback calculations; neither is a complete physical particle or fine-structure prediction.

The beta coefficients in the feedback calculation are obtained from the actual internal-current traces with the intended chiral field multiplicities. This assumes that those are the surviving continuum fields, and it does not count or remove spatial regulator valleys or doublers. The minimal-census root and the supersymmetric proxy are different scale designs, not two calibrations of one proved physical theory. The absence of a positive solution for the minimal census with

$\beta = 4$ is a mathematical inconsistency of that specified combination; it is not an exclusion of other admitted OPH designs.

The three increasing coefficients returned by these closed conditional maps are dimensionless singular coefficients $m_i/v$.

| Scale design | $m_1/v$ | $m_2/v$ | $m_3/v$ |
|---|---|---|---|
| Supersymmetric proxy, $\beta = 4$ | $2.07203042\times10^{-6}$ | $4.28430204\times10^{-4}$ | $7.20520710\times10^{-3}$ |
| Minimal census, $\beta = 7$ | $2.07103435\times10^{-6}$ | $4.28253912\times10^{-4}$ | $7.20232110\times10^{-3}$ |

The interval certificate encloses these values and both admissible memory inverses over each complete root box, so the table and the controller refer to the same conditional source coordinates.

# 5 Localizing the same current and mass operators

## 5.1 Topology, total energy and the radial source

The seed of the defect hypothesis is a precise finite frustration statement. For a finite connected graph, abelian edge offsets $b_e$ and positive weights $w_e$, define

$$\Phi_b(x) = \sum_{e:u\to v} w_e \, \mathbf{1}[x_v - x_u \neq b_e].$$

If a cycle has nonzero signed offset sum, then $\min_x \Phi_b(x) > 0$: the possible costs form a finite set of positive-weight subset sums, and a zero cost would solve every edge equation and force zero cycle sums [3]. This is frustration at fixed compatibility data. Protection of $b$ against an update that changes its class, localization of its energy and a rest mass all need more.

For a one-cochain $b$ on the full carrier complex, face curvature is $F = \delta b$ and $b(\partial c) = F(c)$ for every two-chain $c$. The icosahedral boundary has trivial first homology, and the native load difference $b_{uv} = n_v - n_u$ is exact, so $F = 0$ and all loop sums vanish. An actual defect invariant therefore requires additional transport or field data.

Consider the two-spin cubic hopping operator with forward link $i\sigma_j/2$ and its Hermitian reverse. A declared radial onsite texture is

$$B_x = M \sum_{j=1}^{3} (-1)^{x_j} \frac{x_j}{|x|} \sigma_j, \qquad B_0 = 0.$$

On the odd radius-one cube at $M = 64$, this literal source has exactly two zero modes and a complementary absolute-energy gap at least 62. The normalized kernel frame $W$ has scalar local density $W_x^\dagger W_x = \rho_x I_2$, with $\sum_x \rho_x = 1$. All 28 retained winding matrices have exact spectral enclosures with every multiplicity and source-matrix rounding error controlled. For every retained odd-degree cube, inversion and complete spectral exhaustion prove exactly two zero modes, including weak-mass cases. Degree-three six-level windows hold two zeros and four nonzero levels, which is no count of generations. A complete-Fock contact theorem admits 34 of 84 supplied contact cases, with a ground pair and unique neighboring charged states.

A radial mass profile also needs a total energy law. In the selected ordered-spin rotation orbit, each admissible charged shell costs at least $\pi J\kappa|q|$ with $\kappa > 0$. A hedgehog retaining nonzero degree on arbitrarily many disjoint shells therefore has infinite excess energy in that ungauged identification, and a fermion bound-mode gap omits this background cost. Screening, compact opposite-charge textures or another specified finite-energy structure are substantive dynamical choices. The compact pair below provides the second kind.

### 5.2 One local carrier for defect states, currents and mass coefficients

The radial defect operator admits a direct realization of the native matter currents and a prescribed family mass operator on the same localized modes. The construction is a choice of simulator carrier and coherent pairing service. Its finite operator identity is exact; the choice is not claimed to follow uniquely from overlap consistency.

Let $h$ be the radial operator on $\ell^2(\Lambda)\otimes\mathbb{C}^2$, with its normalized kernel frame $W:\mathbb{C}^2\to\ker h$. Use the strong texture domain, in which the kernel has dimension two and the complementary spectrum has absolute energy at least $d>0$. With

$$J=i\sigma_2,\qquad D_x=(-1)^{x_1+x_2+x_3},\qquad C=DJ,$$

the source identities are

$$C^T=-C,\quad C^\dagger C=I,\quad hC=-C\overline{h},\qquad hW=0,\quad W^\dagger W=I_2,\quad C\overline{W}=WJ.$$

The last identity fixes the quaternion frame through evaluation at the origin and its scalar normalization Gram matrix.

Choose the canonical anticommutation relation algebra on $\ell^2(\Lambda)\otimes\mathbb{C}^2\otimes M_{15}\otimes\mathbb{C}^3$. Here $M_{15}$ is the native matter representation, transported by its actual source intertwiner; the three family copies are declared inputs. For a symmetric all-left Higgs kernel $K\in M_{45}(\mathbb{C})$, define

$$\Delta=C\otimes\overline{K},\qquad B=\begin{pmatrix}h\otimes I & \Delta\\ \Delta^\dagger & -\overline{h}\otimes I\end{pmatrix},\qquad E=\operatorname{diag}(W\otimes I,\overline{W}\otimes I).$$

The antisymmetry of $\Delta$ makes it a valid local creation pairing. Multiplication gives the exact isometric intertwiner

$$BE=EB_0,\qquad B_0=\begin{pmatrix}0 & J\otimes\overline{K}\\ (J\otimes\overline{K})^\dagger & 0\end{pmatrix}. \tag{11}$$

Every singular value of $K$ therefore appears with both signs and twice the multiplicity in the localized Nambu operator, including zero values. Nambu coordinates duplicate annihilation and creation operators; they introduce no physical mirror carrier.

For every actual anti-Hermitian native current $G$, the map $W\otimes I$ intertwines $I\otimes G$ with $I_2\otimes G$. In Nambu coordinates the generator is $\operatorname{diag}(G,\overline{G})$; Hermitian charges use $\operatorname{diag}(Q,-\overline{Q})$. The Higgs background transforms with matter under the uniform internal gauge action. A fixed broken vacuum does not make every gauge generator commute with its mass operator. The resulting map of canonical anticommutation relation algebras intertwines quadratic Heisenberg evolution. On even states, tensoring with the normalized trace on the orthogonal complement gives an exact state inclusion with partial-trace recovery. This complement state is not a vacuum.

Use the positive family matrix $M(P)$ supplied by the memory construction, whose eigenvalues are the retained charged-lepton coefficients $m_i(P)/v(P)$. Choosing $H=(0,v(P)/\sqrt{2})$ and $Y_e=\sqrt{2}M(P)$ realizes those coefficients in $K/v(P)$ itself. The coherent pairing service implements the stationary-mean memory coefficient times its declared gain. Equation (11) does not identify the unread finite memory channel with this Hamiltonian. The calibration provenance of the retained family law is unchanged.

An explicit record service realizes the same Hamiltonian in a controlled weak-step limit. Apply the native current probes to fresh auxiliary legs in their invariant preparations, so their complete twelve-state memory transition is independent of the spatial field. For the updated memory port $j$ choose the field pulse generated by

$$H_r=H_0+\mathbf{1}_C(r)\,gL(P_j),$$

where $H_0$ contains the radial kinetic term and the chosen quark pairing, $L$ is the local lepton pairing map, and $g$ is the source gain. Every $H_r$ has the same core intertwiner. Linearity gives stationary mean $\overline{H} = H_0 + L(gZ_C)$, precisely the Hamiltonian above. If $\|H_r\| \le M$ on the fixed finite Fock space, the reset contraction of the complete memory law and a second-order unitary expansion give

$$\left\|\rho_N - e^{-iT\overline{H}}\rho_0 e^{iT\overline{H}}\right\|_1 \le \frac{4MT}{N(1-p)} + \left(\frac{8}{1-p} + 4\right)\frac{M^2T^2}{N}.$$

The bound includes arbitrary finite untouched references and initial memory correlations; stationary product memory removes the first term. For the displayed cube, $M = 2430\cdot 66 + 1215(1+g)$ suffices. This chosen coherent service uses every record. Its finite unread channel need not be unitary, and its bound is not uniform in volume.

For a complete finite witness, choose the retained quark design $Y_u = \mathrm{diag}(6^{-6}, 6^{-3}, 1)$ and $Y_d = \mathrm{diag}(6^{-6}, 6^{-4}, 6^{-2})$, and set the three minimal neutrino coefficients to zero. Their exponents are design inputs, not topological index conclusions. On the radius-one cube there are 27 sites, 2430 physical annihilator modes and 4860 Nambu rows. The localized sector has 90 annihilators and 180 Nambu rows: 84 positive, 84 negative and 12 zero rows. These numbers count operator degrees of freedom, not observed species. In particular the zero rows do not establish a relativistic massless neutrino sector.

The full complement is retained. Indeed, with $T = \mathrm{diag}(I, C^\dagger \otimes I)$,

$$T^\dagger B T = \begin{pmatrix} h\otimes I & I\otimes \overline{K} \\ I\otimes K & h\otimes I \end{pmatrix}.$$

Its energies are the sums $\lambda(h) \pm s_j(K)$, not a relativistic square-root dispersion. If $\|K\| \le M$ and $2M < d$, the localized band lies in $[-M, M]$ and the complement has absolute energy at least $d - M$. This is single-quasiparticle isolation, not isolation of all many-body energy sums. In the displayed design, choosing $v(P)$ as energy unit, radial hopping one and texture coefficient 64 admits $d = 62$ and $M = 1$.

The common spatial density follows from $W_x^\dagger W_x = \rho_x I_2$; inversion gives mean position zero. A Fourier transform supplies a momentum distribution, without making the localized state a momentum eigenstate. Occupation of these modes does not create the supplied texture winding. Original site-resolved repair can leak out of the kernel, and interactions can mix the two factors. Thus the exact common finite operator supplies neither defect formation, physical rest masses nor an interacting relativistic continuum.

The certificate retains all twelve native current operators, and its complete finite controls include an unstaggered-pairing counterexample.

### 5.3 Compact opposite-degree textures and their isolated modes

A finite-energy texture can carry opposite local winding numbers while approaching a constant exterior. On the two-spin cubic hopping carrier, choose an integer $a \ge 1$, write $R = a + 1$, and define

$$f(x,y,z) = \begin{cases} (x, y, z^2 - a^2), & \max(|x|,|y|,|z|) \le R, \\ (0,0,1), & \text{otherwise.} \end{cases}$$

Set $n = f/|f|$ away from $c_\pm = (0, 0, \pm a)$ and $n(c_\pm) = 0$. The Hamiltonian is $h = T + Mn\cdot\sigma$, where the nearest-neighbor hopping has $\|T\| \le \sqrt{3} < 2$ and $M \ge 8$. The texture and its amplitude zeros are explicit design inputs. The complete unit cubic shells about $c_+$ and $c_-$ have degrees $+1$ and $-1$ under their normalized barycentric interpolation; the exterior shell is constant. Exact rational

regular-value counts check every oriented triangle. This degree classifies the interpolated texture and is not a fermionic superselection charge.

For the selected ordered-spin reference, fill the two core directions by the exterior direction. Each rotated-ground source bond has excess energy in $[0, J]$. With $s = 2a + 3$, precisely $3s^2(s + 1)$ bonds touch the support cube, so

$$0 \le \Delta E_{\mathrm{spin}} \le 3s^2(s + 1)J.$$

The two zero fermion masses and the spin-core fill are distinct declared parts of the construction. The second-quantized Hamiltonian differs from that of the uniform texture by an operator of norm at most $2M(s^3 - 1)$. Its vacuum-energy difference has the same bound, independently of exterior volume. These are total texture-energy bounds; they do not assign a constituent rest mass to a localized mode.

Projection onto the two core sites gives $h = \left(\begin{smallmatrix} 0 & B^\dagger \\ B & D \end{smallmatrix}\right)$. The exterior block has no spectrum in $(-M + 2, M - 2)$, and $\|B\|^2 = b^2$, where $b^2 = 7/4$ for $a = 1$ and $b^2 = 3/2$ otherwise. There are exactly four low eigenvalues. Their Feshbach equations give

$$|E| \le r := \frac{b^2}{M - 4}, \qquad \frac{\|\psi_{\mathrm{exterior}}\|^2}{\|\psi_{\mathrm{cores}}\|^2} \le \frac{b^2}{(M - 2 - r)^2}.$$

These bounds are uniform in volume. For a cube of radius $L \ge R + 1$, resolvent path expansion and monotone Feshbach eigenvalue crossings give

$$|E_j(L) - E_j(\infty)| \le \frac{2b^2}{M - r - 2} \left(\frac{2}{M - r}\right)^{2(L+1-R)}.$$

Thus the infinite lattice also has precisely four isolated modes in $[-2, 2]$. At $a = 1$, $M = 64$, $L = 3$, complete rational interval certification of all 686 eigenvalues excludes zero from all four low levels and from their infinite-volume limits. The numerical levels are approximately $\pm 0.003905899328$ and $\pm 0.003906437151$. The finite-volume filled-sea energy difference is approximately 127.9700 in the same hopping units.

Let $W$ span this four-dimensional band, $A = W^\dagger h W$, and $C = D_{\mathrm{stagger}} i\sigma_2$. Particle-hole symmetry gives $C\overline{W} = WS$ with $S = W^\dagger C \overline{W}$ antisymmetric and unitary. The actual twelve native current operators intertwine through $W \otimes I_{45}$. The same local pairing $\Delta = C \otimes \overline{K}$ has the exact core

$$\begin{pmatrix} A \otimes I & S \otimes \overline{K} \\ (S \otimes \overline{K})^\dagger & -\overline{A} \otimes I \end{pmatrix}.$$

Its energies are those of $A$ plus or minus the singular values of $K$. Consequently the compact texture shifts the retained family coefficients; dropping $A$ gives an incorrect operator inclusion. A local Pauli perturbation at the two cores preserves every shell datum while changing these energies. Winding alone therefore does not pin a zero mode for this carrier. The construction supplies a compact background and controlled localized spectrum, without autonomous formation, stability, a relativistic mass shell or physical particle identification.

### 5.4 Protected texture records and controlled band transport

A declared local repair law protects the compact pair's two winding charges. Let $f_x$ be its unnormalized reference texture and fix the zero cores and constant exterior. At every noncore site retain a rational vector $v_x$ with $\|v_x - f_x\| \le \eta \|f_x\|$, $0 < \eta < 1/4$, using $v_x/\|v_x\|$ in the actual Pauli potential.

A one-vertex proposal is accepted only if it satisfies this budget and has nonnegative scalar product with the old vector and every other vertex of every incident shell triangle. Shared-face data are checked on both shells. Rejected proposals remain recorded identity operations.

**Proposition 5.1** (Guarded winding and uniform isolated band)**.** *Every admitted finite history preserves the two shell degrees. Its spatial Hamiltonian has exactly four isolated modes with* $|E| \le r = b^2/(M-4)$ *and complementary spectrum outside* $(-g, g)$*,* $g = M-2$*, where* $b^2 = 7/4$ *for adjacent unit shells and* $b^2 = 3/2$ *for separated shells. For two admitted band projectors* $\Pi_v, \Pi_w$*,*

$$\|\Pi_w - \Pi_v\| \le \beta = \frac{4M\eta c}{d^2}, \qquad c = \frac{g+r}{2}, \quad d = \frac{g-r}{2}.$$

*The statements are uniform in the exterior volume.*

*Proof.* During an accepted replacement, every interpolated shell vector is a convex combination of nonzero vectors with pairwise nonnegative scalar products. Its squared norm is at least $\sum_i t_i^2 \|v_i\|^2 > 0$, so normalization gives a shell homotopy. Every noncore onsite eigenvalue stays $\pm M$ and the four core coordinates stay zero; the compact-source Schur proof therefore gives the same band count and localization bounds. Normalization gives $\|H_w - H_v\| \le 4M\eta$. On $|z| = c$ both resolvents have norm at most $1/d$. Their identity and the contour formula for the two projectors give the displayed bound. □

For $\beta < 1$, define

$$S = \Pi_w\Pi_v + (1-\Pi_w)(1-\Pi_v), \qquad U = S(S^*S)^{-1/2}, \qquad S^*S = 1 - (\Pi_w - \Pi_v)^2.$$

Since $S\Pi_v = \Pi_w S$ and the positive factor commutes with $\Pi_v$, $U$ is unitary and $U\Pi_v U^* = \Pi_w$. The literal particle-hole relation $H_v C = -C\overline{H_v}$ implies $\Pi_v C = C\overline{\Pi_v}$, and the polar formula gives $UC = C\overline{U}$, hence $UCU^T = C$. Thus $U \otimes I_{45}$ transports all twelve uniform native currents and the same pairing tensor $C \otimes \overline{K}$. For $W' = UW$, the pairing matrix $W'^* C \overline{W'}$ is unchanged, while $A_w = W'^* H_w W'$ retains the actual changed spatial energies. Second quantization also transports $N_v = d\Gamma(\Pi_v \otimes I_{45})$ and $\lambda(N_v - 90)^2$.

The explicit source uses $M = 64$, $\eta = 1/512$ and four equiprobable proposals at each record: two opposite transverse changes at the shared-face center, one forbidden zero and one forbidden core change. Its three texture states have transition rows $(1/2, 1/4, 1/4)$, $(0, 3/4, 1/4)$ and $(0, 1/4, 3/4)$. All twelve state/proposal records are retained. Each branch applies its texture transition and $\sqrt{q_r}\,\Gamma(U_r)$, with identity control for a repeated texture, so the complete instrument preserves trace. The band is transported exactly at record endpoints. Omitting $U_r$ instead gives leakage probability at most $\min(1, n^2\beta^2)$ after $n$ quenches and intervening source evolution; the numerical controls exhibit nonzero leakage.

A nearest-neighbor snake through the $N = 2(2L+1)^3$ spatial-spin coordinates compiles each $U$ by adjacent Givens rotations: at most $N(N-1)/2$ two-mode gates and $N$ onsite phases. The complete $N = 686$ controls replay 234,955 two-mode gates per nontrivial texture pair. Addressed quadratic pulses, their timing and drift cancellation are supplied services; this is not a compilation into unchanged C/E repair. Intermediate pulses need not preserve the band, and the controller supplies the endpoint energy change, bounded by $2r$ on the one-body band. Erasing the guards permits zero-degree textures with the same four-mode count. The result establishes controlled protection and transport on this law, without autonomous core motion, local Gauss dressing, physical species identification or relativistic dispersion.

### 5.5 Gauge-covariant paired compact bands

The compact source extends to supplied classical unitary internal links on its 343-site radius-three lattice. The gauge paper [5] proves that with onsite texture amplitude 64 and pairing of norm at most one, the complete Bogoliubov–de Gennes operator keeps a separated band whose particle grading defines 180 canonical annihilation modes and an integer occupation $N_r$. The quartic $\lambda(N_r - 90)^2$ of Section 5.16 is therefore well-defined on every link background and equals the source compact interaction at trivial links. Transforming link and pairing together preserves band and interaction covariantly, and declared polar controllers transport the grading between records.

The same paper gives the separate native quantum-link design: a 60-state link on the five matter blocks with exact local non-Abelian Gauss constraints on the full Fock space, and a conditional rotor completion that leaves five link-energy differences unselected. A literal native two-site raising-link example has a noninteger naive occupation even on a Gauss singlet, so the canonical band above does not transfer to operator-valued links by substitution. A dynamical quantum-link realization of the interacting compact band is not established here.

### 5.6 Native record response

The fixed biased binary source also supplies an observable response without an attached field action. Give ports $0, 1$ weight 25 and the other ten weight 8, select each of the thirty seams uniformly, and tag current and contrast repairs with equal probability. Let $P$ be its column-stochastic occupation transition and $\pi(x) \propto \prod_a \theta_a^{n_a(x)}$ its stationary law. On every complete record define $Y_e = 60\,\mathbf{1}[\text{contrast law, seam } e](n_a - n_b)$ for $e = (a, b)$, with zero score on reserved addresses. This uses the reported incoming contrast. If $B_e$ is the transition weighted by that score, then

$$\mathbf{1}^T B_e = d_e^T, \qquad B_e \pi = \operatorname{diag}(\pi) g_e, \qquad g_e = \frac{\theta_a - \theta_b}{\theta_a + \theta_b} \mathbf{1}[n_a + n_b = 1].$$

The second identity includes the measuring repair's backaction. With rows $d_e^T$ and $g_e^T$ collected in $D, G$, and $m = D\pi$, the ordered stationary covariance and its discrete-attempt resolvent are

$$C(k) = DP^{k-1}A \quad (k \geq 1), \qquad A = \operatorname{diag}(\pi) G^T - \pi m^T,$$
$$\sum_{k \geq 1} z^{-k} C(k) = D(zI - P)^{-1} A.$$

All 22 equal-weight seams have zero columns in $C(k)$, despite nonzero same-time shot noise. A passive occupation correlator omits this backaction. Exact ranks 66 for $[\mathbf{1}^T; D; \ldots; DP^{12}]$ and 65 for $[A, PA, \ldots, P^{10}A]$, together with the squarefree characteristic polynomial, show that all 65 nonstationary population relaxation poles appear. They are positive real decay factors, without a particle-mass interpretation. The biased occupation means require pair correlations: adjoining one-step images raises their span from 12 to 19.

The calculation retains all 721 interpreter slots and arbitrary untouched references. Complete record probabilities observe only diagonal source blocks, so the rank statement is population observability, not quantum-state tomography. After 1600 unread preparation attempts, the reduced source/reference half trace distance is at most $\varepsilon = 542159/10^{12}$; any fixed future $r$-fold score moment has error at most $2\,60^r \varepsilon$. The preparation archive can remain correlated. No finite sampling or stable pole-inversion bound is asserted (`model/native_record_response`, [1]).

**Positive transfer from complete records.** Write $\Pi = \mathrm{diag}(\pi)$. Detailed balance gives the symmetric transfer $S = \Pi^{-1/2} P \Pi^{1/2}$, and the actual minimum diagonal $\min_x P_{xx} = 791/990$ gives

$$\frac{296}{495} I \le S \le I, \qquad H_{\mathrm{tr}} = -\log S \ge 0.$$

For each raw flag the native reversal satisfies $T_{\bar{h}} = \Pi T_h^T \Pi^{-1}$. It reverses word order and exchanges incoming and outgoing contrast using the retained current. If $F$ is any scalar future-record cylinder and $f_F$ its conditional expectation on the initial configuration, its reflected form across $n$ unread attempts is $\langle f_F, (P^T)^n f_F \rangle_\pi \ge 0$. The preceding observability rank makes the reflection quotient exactly 66-dimensional. Centered contrasts therefore have the positive reflected spectral representation

$$D P^n \Pi D^T - m m^T = \sum_{j=1}^{65} \lambda_j^n v_j v_j^\dagger, \qquad v_j \ne 0.$$

The ordinary oriented covariance $C(1)$ is different: the centered score $Y_{(0,5)} - Y_{(0,1)}$ has the naive unreflected quadratic form $-83481968/1858725825$. Reflection must use the actual record dictionary.

This is a finite Euclidean reconstruction from scalar records. Spectral calculus defines a unitary continuation but supplies no native protocol executing it. Its pure ground direction $\sqrt{\pi}$ is not the native faithful working state, and $H_{\mathrm{tr}}$ differs from that state's modular generator $K = -\log \Pi$. For $x = \{0,1\}$ and $y = \{1,5\}$ the exact element $[S,K]_{yx} = \sqrt{2}\log(8/25)/99$ is nonzero; hence $[H_{\mathrm{tr}}, K] \ne 0$. The two clocks cannot be scalar rescalings and shifts of one another. Physical time, spatial locality, operator-valued history positivity and particle masses require additional maps (`model/native_record_response/TRANSFER.md`, [1]).

**Operational clock comparison.** The distinction has a finite operational witness. For the accessible $Y$ probe on seam $(1,7)$, the working state $\Pi$ has zero mean under every modular-clock time. The proposed identity identification with $e^{-itH_{\mathrm{tr}}}$ instead gives

$$f(t) = \mathrm{Tr}(Y e^{-itH_{\mathrm{tr}}} \Pi e^{itH_{\mathrm{tr}}}), \qquad f'(0) > \frac{7}{1000}, \qquad f(1/10) > \frac{1}{2000}.$$

A rational degree-ten logarithm enclosure proves the derivative bound. Reality makes $f$ odd, and $\|H_{\mathrm{tr}}\| < 103/200$ controls the cubic remainder. The native preparation and complete analyzer bounds leave an output half-trace separation greater than $1/5000$ if both proposed clock services have half-diamond error at most $10^{-5}$. This rejects their identification on the same working reference, including reference-preserving unitary changes of basis. It neither executes the transfer unitary nor excludes a different state/readout map. Bare repair records, with the final carrier traced, are insensitive to any interleaved diagonal phases; the declared coherent analyzer supplies the phase-sensitive readout (`model/native_clock_comparison`, [1]).

**Complete records for disjoint repairs.** For two fixed native commands with disjoint endpoints, every pair of actual raw Kraus operators commutes on the full 66- or 78-state carrier. Both executions must finish. The comparison permutes command-instance slots with their exported and retained records, and relabels the known program identifiers; their literal digests differ. The resulting complete classical-quantum channels are identical for every initial state and untouched reference. All $25^2$ raw-slot pairs, including holds and zero addresses, are retained. Adjacent swaps extend the identity to fixed schedules related by such disjoint interchanges. The literal chronological tapes need not agree, and the shared queue's implementation dependencies are unchanged. Overlapping

endpoints, selectors using earlier records and memory timeouts give explicit failures. This is a finite equivalence of observation protocols; it supplies no physical event quotient, spatial dimension or light cone (`model/native_record_causality`, [1]).

**Canonical source restriction.** The fair connected federation has $V_\ell = 240\,4^\ell$ ports and a declared global occupation $m$. Its canonical state $\rho_{V,m} = \Pi_m/\binom{V}{m}$, where $\Pi_m$ projects onto total occupation $m$, is preserved by each fixed fair native seam's unread source channel. Tracing to any $v$ ports gives the full number mixture

$$\rho_W = \sum_{k=0}^{v} h(k)\rho_{v,k}, \qquad h(k) = \frac{\binom{v}{k}\binom{V-v}{m-k}}{\binom{V}{m}},$$

with impossible binomial coefficients zero. These restrictions compose exactly for nested regions. Orthogonal number sectors give the half trace distance $D(\rho_W, \rho_{v,q}) = 1 - h(q)$; imposing a sharp local number is a different state. For $p = m/V$ and $\beta_p = \mathrm{diag}(1-p, p)$, sampling with and without replacement gives

$$D(\rho_W, \beta_p^{\otimes v}) \le \min\{1, v(v-1)/(2V)\}, \qquad D(\rho_W, |0\rangle\langle 0|^{\otimes v}) \le \min\{1, vm/V\}.$$

A common complete local instrument contracts these bounds, including its records; a correlated external reference needs a joint input bound. The number law requires $O(v)$ rational operations. All thirteen outcomes of a twelve-port restriction at $\ell = 200$ are evaluated exactly with the 408-bit population integer, without allocating the global state. These are static state comparisons. They select no physical capacity map, occupation trajectory $m(V)$, finite preparation, clock or dynamical restriction. The fair federation is distinct from the biased local member used above (`model/native_capacity_restriction`, [1]).

The existing independent-cell grammar supplies a distinct dynamic extension. Append unused 66- or 78-state factors to any admitted native repair/hold program, preserving its parameters, active cell addresses and record controller. Every raw branch is $K_h \otimes I_B$. Consequently partial trace over $B$ intertwines the complete adaptive instrument for arbitrary correlated inputs and untouched references. The comparison retains chronological records, FIFO exports and matched stopping outcomes, and checks the exact lifted program identifier before relabeling it. These restrictions also compose. Declared allocation and specification queries can distinguish the programs. The host evaluator stops before executing products above 64 cells, so equality of executed histories requires a common admitted and evaluated domain. Compiled coherent source-word services have a separate grammar and are not lifted by this theorem. This proves a spectator-factor extension without connected geometry or interactions. A direct single-attempt comparison with the fair federation fails: the same coin record has probabilities 25/33 and 1/2, giving total variation at least 17/66 (`model/native_capacity_restriction/PRODUCT.md`, [1]).

**Connected weighted references.** The same endpoint rule has a consistent weighted extension on the retained connected federation graph. For positive vertex weights $w_i$, write $e_m(w)$ for the elementary symmetric polynomial and set $\pi_m(S) = \prod_{i\in S} w_i/e_m(w)$ on $|S| = m$. Every seam preserves this reference by conditional heat-bath detailed balance. Its full restriction to $A$ has number weights

$$h_A(k) = \frac{e_k(w_A)e_{m-k}(w_{A^c})}{e_m(w)}.$$

These restrictions compose. Repeating the local weight word gives the H66 reference conditional on two loads in a cell; cross-cell repair changes that local number. The faithful product family

with occupations $p_i = zw_i/(1+zw_i)$ also restricts exactly for fixed fugacity $z$ and inherited weights. The modular transfer $j\log(w_a/w_b)$ is independent of total occupation and graph size. It is not a particle mass.

For a fixed independent seam selector, replacing crossing repairs by a complete hold control changes the $t$-attempt channel by at most $1-(1-\beta)^t$ in half diamond norm, where $\beta$ is its boundary-selection probability. Each cell has thirty internal and three crossing seams. Under the uniform seam selector, through the $k$th internal repair, the no-crossing probability is $(10/11)^k$ at every federation level; increasing exterior capacity does not establish vanishing error in that local clock. Symmetrizing the same population law gives local positive terms

$$h_{ab} = \frac{1}{w_a+w_b}\begin{pmatrix} w_b & -\sqrt{w_a w_b} \\ -\sqrt{w_a w_b} & w_a \end{pmatrix}$$

on $|10\rangle, |01\rangle$, zero on $|00\rangle, |11\rangle$. Their weighted sum annihilates $\sqrt{\pi_m}$. This is a relaxation representation. The weight assignment and fugacity are declared, and the native program admission rule rejects weighted connected federations; no source selection, prepared coherent ground or relativistic limit is inferred (`model/weighted_source_family`, [1]).

**Local weighted state limits.** The repeated native weight word admits a controlled local limit without enumerating its global carrier. Let $B$ be the number of twelve-port cells, declare total occupation $rB$ with $r \in \{2,4\}$, and choose the positive fugacity solving

$$1000z^2+32z-1=0 \quad (r=2), \qquad 800z^2-z-2=0 \quad (r=4).$$

The product reference $Q_B = Q_1^{\otimes B}$ then has mean $rB$, and its conditioning on that total is exactly the canonical reference $P_B$. A binomial-mode bound gives $Q_B(S=rB) \geq 1/(240B)$. Cell exchangeability and the finite entropy chain rule distribute this conditioning cost: for every set of $k$ cells, including $k$ that do not divide $B$,

$$D(P_k\|Q_1^{\otimes k}) \leq \frac{k\log(240B)}{B}, \qquad \tfrac{1}{2}\|P_k - Q_1^{\otimes k}\|_1 \leq \min\left\{1, \sqrt{\frac{k\log(240B)}{2B}}\right\}.$$

Here $D$ is classical relative entropy, the second inequality is finite Pinsker, and every local number sector is retained. A common trace-preserving local instrument contracts the bound; a numerical map with bounded gain requires that gain in the comparison. The instrument must act on every retained local number sector: a two-load coherent service does not automatically extend to this mixture. Correlated references require a joint input-state bound. For $B = 20\,4^{200}$ the one-cell bound is below $2\,10^{-60}$, computed with integer and rational enclosures. Matching mean occupation two does not confine a cell to its two-load sector. The theorem selects neither density nor physical capacity and supplies no preparation time or connected dynamical limit (`model/weighted_local_limit`, [1]).

**Local records and finite-volume response.** For a fixed repair schedule, a local observation consists of a final port region and selected complete local records at their original event positions. An untouched reference is retained; remote record values and the final exterior are traced. Reverse scanning the schedule adds the endpoints of each retained read and each unread command touching the accumulated support. Every disjoint unread complete channel cancels under partial trace. Thus the complete observed instrument factors exactly through the initial state on this dependency cone

and its reference, including initial coherences and correlations. State-independent random selectors can be averaged afterward. Record-adaptive selectors and intrinsic queue halts require separate analysis; the full global archive is a different observable.

Compare with a finite port region whose crossing and exterior commands act as holds, retaining the original selector and clock. Coupling the selector tapes bounds half the diamond distance of the two local output channels by the probability that the dependency cone exits the region. An initial joint state error adds to this bound. For independent seam choices, root set $S$, maximum degree $d$, maximum seam probability $q_*$ and graph-distance margin $r$, a union over ordered paths gives the sufficient bound

$$p_{\mathrm{exit}} \le \min\{1, |S|\tbinom{T}{r+1}(dq_*)^{r+1}\}.$$

The roots include both endpoints of every retained state-dependent read. This conservative cone can exceed the support obtained from the precise record times.

For independent uniform seam selection and repair probability one, on the actual fair connected source at levels zero through two, each cell has thirty internal seams and three distinct exterior neighbors, each with five seams leaving the one-neighbor halo. For a whole-cell root with retained internal-cell records, this reduces cone exit to four transient states $n = 0, 1, 2, 3$, with $n$ counting the halo ports reached. With $M = 630\,4^{\ell}$ seams, the only nonzero entries are

$$Q_{nn} = 1 - \frac{3+4n}{M}, \qquad Q_{n+1,n} = \frac{3-n}{M}, \qquad p_{\mathrm{exit}} = 1 - \mathbf{1}^{\top} Q^{T} e_0 \le \min\{1, 15\tbinom{T}{2}/M^2\}.$$

Here $Q^T$ denotes the $T$th power. At level zero and $T = 30$ this probability is below 0.015. The combinatorial attachment below proves the halo geometry for every returned source routing. Stopping immediately after the $k$th internal repair instead gives

$$R = 30(D - B_0)^{-1}, \qquad D_{nn} = 33 + 4n, \quad (B_0)_{n+1,n} = 3 - n, \qquad p_{\mathrm{exit}} = 1 - \mathbf{1}^{\top} R^{k} e_0.$$

The elapsed global time is retained without conditioning. The exit probability is independent of $M$, equals $677/50061$ at $k = 1$, and is at most $\min\{1, k(k+1)/120\}$. For bounded scores $|f| \le 1$ and an analysis parameter $0 \le q < 1$, the normalized Abel response error is at most

$$2\left[1 - (1-q)\mathbf{1}^{\top}(I - qR)^{-1} e_0\right] \le \min\{2, q/[30(1-q)^2]\}.$$

At $q = 1/2$ the sharper value is $151/3432$. These are complete local-record response bounds, with no pole-energy interpretation. The weighted connected extension has the same conditional locality construction but no native admission; its reference is not assigned to the fair source. The proof and bounded interpreter checks supply neither a physical clock nor a relativistic limit (`model/native_local_response`, [1]).

**Occupation-sector readout.** On any connected binary source with $V$ ports, the conserved total $m$ and complete records of $V - 1$ fixed-seam repairs along a spanning tree determine the initial configuration. If $s_i$ is the accumulated recorded displacement, the incoming contrast on $(a, b)$ gives the initial difference $d - s_a + s_b$. Propagate these differences to offsets $o_i$; the common offset is $(m - \sum_i o_i)/V$. Replaying the currents also determines the terminal configuration. Every nonzero complete word therefore has the map $\sqrt{w_h}|y_h\rangle\langle x_h|$, including arbitrary untouched references, and coarse graining by $x_h$ gives the configuration projectors. Without a supplied total, the empty and full configurations share one rank-two branch that preserves their coherence. All other supported words decode a unique configuration. This ambiguity cannot be removed by record-adaptive repair selectors.

The actual connected members at levels zero through two admit trees with 239, 959 and 3839 commands. An exact lazy query of a complete raw word uses the original continuation kernel and retains its archive; it does not turn the expanded evaluator's resource stop into a completed execution. A final tree read permits reverse-current reconstruction of a preceding complete repair history. Its terminal basis state is outcome-labelled. For any controller of at most $T$ repairs, positive placement probabilities retain a no-motion event from every basis input with probability at least $(1 - r + rp_{\min})^T$, where $r$ is the repair probability. On a nontrivial fixed-number sector, this obstructs exact deterministic reset of every input to one target by bounded repair-only control. It permits approximate or heralded preparation and the separately declared coherent services (`model/native_sector_readout`, [1]).

**Native correlation sectors.** For the fair connected source with $M$ seams, the population transfer is $P_q = I - \sum_{ab}(I - S_{ab})/(2M)$ on the $q$-occupation sector. Let $U$ add a site to a subset, $D = U^*$ and $W_r = \ker D_r$. The standard harmonic decomposition of the Boolean slice [27] follows from $DU - UD = (V - 2r)I$ on degree $r$. Its source-preserved degree-$r$ subspace has dimension $\binom{V}{r} - \binom{V}{r-1}$, for $r \le \min(q, V-q)$, and is unitarily equivalent to $P_r|_{W_r}$. This is a decomposition of the unread transfer, with no assertion that each recorded quantum branch preserves these subspaces.

For two disjoint native seams $(a, b), (c, d)$, put $h = e_{ac} - e_{ad} - e_{bc} + e_{bd}$ and $\phi = (n_a - n_b)(n_c - n_d)$. Then $D_2 h = 0$, and in the uniform fixed-number reference

$$v_q = \mathbb{E}\phi^2 = 4\frac{\binom{V-4}{q-2}}{\binom{V}{q}}, \qquad \frac{\mathbb{E}[\phi(X_0)\phi(X_t)]}{v_q} = \frac{h^* P_2^t h}{4}, \qquad 2 \le q \le V - 2.$$

An actual four-read experiment realizes this curve: read both seams, score their outgoing contrasts $(d_1 - 2j_1)(d_2 - 2j_2)$, run $t$ uniform attempts, then read both seams and score the incoming contrasts $d_3 d_4$. The complete fair instruments are unital; the first scored output is $\phi\tau_q$ and the last scored effect is $\phi$. Thus their joint score equals the displayed covariance with measurement backaction included. Every raw flag is retained. Using incoming contrasts at both ends changes the experiment and fails this identity. A preparation error $\epsilon$ in half trace distance gives normalized error at most $2\epsilon/v_q$; $v_q$ decreases as $V^{-2}$ at fixed $q$. The normalized poles are independent of occupation on a fixed graph, not of graph size or clock. Different correlation degrees may share poles. Neither degree nor pole is assigned a particle identity (`model/native_correlation_sectors`, [1]).

**Preparing native correlation records.** The stationary response can be prepared at the level of its complete signed score from an arbitrary quantum input in the same fixed-number sector. For possibly different past and future seam contrasts $\phi_i, \phi_j$, let $H_e = (I + S_e)/2$ on occupation functions. Pulling the entire four-read experiment back to its initial cut gives the diagonal effect

$$A_{ij,t} = H_{e_i} H_{f_i}[\phi_i P_q^t \phi_j], \qquad \|A_{ij,t} - \mathbb{E}_{\tau_q} A_{ij,t}\|_2 \le \sqrt{v_q}.$$

The two past seam averages include measurement backaction. This effect has squarefree degree at most four. Put $s = \min(4, q, V - q)$ and $N_s = \binom{V}{s} - 1$. The mean-zero polynomial space is invariant under all site permutations, even when the graph is not vertex-transitive. Its orthogonal projector therefore has constant diagonal $N_s$ in the uniform $L^2$ normalization. Cauchy–Schwarz and the exclusion gap $\gamma = \lambda_1(L)/(2M)$ give, after $n$ independent fair warmup repairs,

$$\frac{\|Z_R - C_{ij}(t)\rho_R\|_1}{v_q} \le \sqrt{N_s/v_q}\,(1 - \gamma)^n.$$

Here $Z_R$ is the terminal reference operator weighted by the complete score, and $R$ is any untouched reference initially correlated with the source. The diagonal input blocks on $R$ are positive and have total trace one, which proves this extension. Warmup records are summed without conditioning; their full joint law is not asserted to be stationary.

Integer square roots and the bound $(1-\gamma_0)^{\lceil 1/\gamma_0 \rceil} \le 1/2$ give sufficient attempt budgets from a rational gap lower bound. On the fair 66-state source, 352 warmup repairs followed by four reads and two lag attempts give normalized error at most 33/65536; exact propagation of all input columns gives a tighter rational enclosure. At fixed density the prefactor grows polynomially rather than with the full sector dimension. The gap and sample cost retain their capacity dependence. All raw records and host stops are retained; assigning score zero to stopped runs adds $p_{\mathrm{stop}}/v_q$ to the bound. Single-program budgets retain the original $2^{32}$-attempt limit. The same argument applies to any separately implemented degree-four scored effect using its exact centered norm. It supplies neither full-density preparation nor a physical vacuum (`model/native_correlation_preparation`, [1]).

**Native spatial motion and clocks.** Midpoint subdivision preserves a simple closed triangular complex; each cell has three distinct neighbors. The source assigns their crossing edges to distinct ports, so each of the three halo ports has five seams leaving the halo and none joining another halo port. This proves the four-state halo premise at every combinatorial level for every returned routing. It does not assert successful floating-point construction or host execution at arbitrary levels.

In the existing auxiliary affine cell-centroid dictionary, normalized to unit circumsphere, all ports of a cell share one coordinate. Writing $n = 2^\ell$, there are $30(n^2-n)$ within-face and $30n$ original-face hinge crossings. Their exact total squared length is

$$E_\ell = 20 - 4\sqrt{5} + \frac{-40+16\sqrt{5}}{3\,2^\ell} < 12.$$

For a stationary uniform single-unit source, $P = I - L/(2M)$ with $M = 630\,4^\ell$ seams and $V = 240\,4^\ell$ ports. Positivity gives $I - P^T \le T(I-P)$ and therefore

$$\mathbb{E}\|x(X_T) - x(X_0)\|^2 \le \frac{TE_\ell}{MV} \le \frac{T}{12600\,16^\ell}.$$

Markov's inequality gives a displacement-probability bound by dividing by the squared threshold. At $T = \lfloor 1260\,8^\ell t \rfloor$, this mean square is at most $t/(10\,2^\ell)$. It is a finite stationary path bound in comparison coordinates, with no physical light-speed assignment.

For a fixed set of $d$ tick seams, a completed tick consists of a geometric number of remote repairs followed by one incident repair. Both conditional transfers preserve the uniform state. The squared triangle inequality and stationarity bound displacement after $k$ completed ticks by $k^2/(10d\,4^\ell)$. Substituting the mean elapsed time into the deterministic-time bound would not justify this result. A finite attempt cap retains timeout as a separate outcome. Along a shortest path of $D$ edges to a receiver, one positive native history of weight $(2M)^{-D}$ arrives at its first incident tick; that tick has no strict port-distance support bound uniform in capacity. Global attempt locality is unchanged. A final tree detector and the body currents reconstruct the recorded path; the prepared law, auxiliary coordinate dictionary and complete archive access are explicit inputs (`model/native_spatial_scaling`, [1]).

**Spatial pair responses.** Use one declared $2 \le q \le V-2$ and its reference $\tau_q$, the same fair graph and the same attempt clock for a menu of disjoint-seam contrast pairs. The complete reflected probes give

$$C_{ij}(T) = \frac{v_q}{4} h_i^* P_2^T h_j, \qquad B_{ij}(T) = C_{ij}(T)/v_q.$$

The matrix $B(T)$ is positive semidefinite; $B(0)$ is the probe Gram matrix, not necessarily the identity. The two-subset coefficients implement the exact intertwiner without changing the working occupation to two. For a radius-$r$ local comparison containing every probe endpoint, maximum degree $\Delta$ gives normalized response error at most $2\min\{1, |S|\binom{T}{r+1}(\Delta/M)^{r+1}\}/v_q$. The compared window retains all local number sectors, and remote raw values are traced at this observation boundary. A global archive-dependent controller is outside that locality bound.

The source also fixes a finite exclusion correction. Compressing the independent two-walker generator to distinct symmetric pairs gives $G_{\mathrm{ind}} = G_2 + 2A$, where $A$ projects onto adjacent pairs. Its collision-retaining control has surviving transfer $R = P_2 - A/M$ and an absorbing collision flag. The identity

$$P_2^T - R^T = \sum_{k=0}^{T-1} P_2^{T-1-k}(A/M)R^k$$

is entrywise nonnegative. Its column sum is the complete collision probability, bounding the normalized probe-response difference under the initial coefficient law $|h_j|/4$. No survivor renormalization occurs. On a four-port path, the one-attempt discrepancy is $1/12$; matching the one-body transfer therefore does not determine the pair response.

For bounded macroscopic profiles, subtracting the constant and one-body components of a product of native density fields gives an exact primitive degree-two field on the same $q$ slice. With $V_\ell = 240\,4^\ell$, $q_\ell = V_\ell/3$ and $\lfloor 1260\,16^\ell \tau \rfloor$ attempts, along the existing native density subsequence, these fields converge to $W_\tau(f,g) = {:}Y_\tau(f)Y_\tau(g){:}$, with

$$\mathrm{Cov}(W_\tau(f,g), W_u(h,k)) = K_{\tau,u}(f,h)K_{\tau,u}(g,k) + K_{\tau,u}(f,k)K_{\tau,u}(g,h).$$

Here $Y$ is the inherited Gaussian density field and $K$ its heat covariance. Canonical exponential-moment bounds give uniform eighth moments and justify the product and covariance limits. A final native spanning-tree detector and reverse-current readback access these occupation snapshots; this global observation is distinct from the local four-read menu. The result is the quadratic sector of a Gaussian diffusive field. The finite exclusion correction does not by itself establish an interacting relativistic continuum (`model/native_pair_response`, [1]).

**Native attempt time and the local exchange action.** The same source fixes $G_q = \sum_e (I - S_e)$ and $P_q = I - G_q/(2M)$. Its bound $10I/21 \le P_q \le I$ gives, by scalar spectral calculus,

$$\left\| P_q^n - e^{-nG_q/(2M)} \right\| \le \frac{3}{5n}, \qquad n \ge 1.$$

Indeed, for $p \ge 10/21$ and $\delta = 1-p$, $0 \le -\log p - \delta \le 21\delta^2/20$; maximizing $n\delta^2 e^{-n\delta}$ proves the bound. It holds on every occupation sector, independent of its dimension, and bounds each entry of the normalized stationary response matrix. The rescaled time is the exact count $t_n = n/(2M)$, without a replacement Poisson scheduler or a physical time unit.

Generator convergence has a narrower domain. On correlation degrees at most $r$, $\|G_q\| \le 2r\Delta$ and, for $M > r\Delta$,

$$0 \le -2M \log P_q - G_q \le \frac{r^2\Delta^2}{M - r\Delta} I.$$

At half occupation the full-sector generator difference is at least $M/16$, although its endpoint semigroups obey the dimension-independent bound above. Exact sparse moments give rational enclosures of both response functions. Preparation and local-window errors add with their stated normalizations. The Poisson-count representation of this exponential has a different complete history-length law; the endpoint estimate identifies neither that law nor a native unitary implementation. These results fix a finite local exchange response, with no chiral matter, Lorentzian action or physical mass assignment (`model/native_exchange_clock`, [1]).

### 5.7 An observable native contact mode

The fair twelve-port action at working reference $I_{66}/66$ has an observable correlated mode. On unordered distinct pairs write $G_2 = \sum_e (I - S_e)$ and let $W_2 = \ker D$, where $D$ sums pair coefficients at each port. Its dimension is 54. If $Q$ projects onto $W_2$ and $A$ onto adjacent pairs, compression of the independent one-body sum gives $G_2 + 2A$. On $W_2$ this yields

$$H = G_2|_{W_2}, \qquad H_0 = H + 2QAQ \geq 2(5 - \sqrt{5})I.$$

The coefficient two follows from the literal collision loss; it is not fitted. For the adjacent-coordinate insertion $E_A$, put $U = QE_A$ and $R_0(z) = (H_0 - z)^{-1}$. Direct multiplication proves

$$(H - z)^{-1} = R_0 + 2R_0U(I - 2U^*R_0U)^{-1}U^*R_0.$$

Below the stated independent-pair threshold, the middle kernel detects eigenvalues with multiplicity. This is a finite comparison threshold, without an infinite-volume scattering interpretation.

Exact integer characteristic polynomials give a fivefold native eigenvalue $E$ at the lowest root of

$$E^3 - 30E^2 + 280E - 776 = 0, \qquad 5 < E < 51/10.$$

It lies more than $2/5$ below the comparison threshold. The actual seam contrast $(n_0 - n_1)(n_2 - n_4)$ has a twelve-dimensional cyclic primitive space. Its exact scalar resolvent has coprime numerator and denominator, so all twelve distinct primitive poles have positive weight. At this lowest line the normalized weight obeys $1/100 < w_E < 1/50$; decimals are $E \simeq 5.01827276928$ and $w_E \simeq 0.01943798293$. Complete reflected four-E records have raw weight $(2/33)w_E$ and attempt transfer factor $1 - E/60$. All raw branches and measurement disturbance are retained.

The isolated-cell mode does not extend unchanged to the connected source. For any normalized vector in the fifteen-dimensional sum of the cubic eigenspaces, every three-port boundary attachment satisfies

$$\|(I - P_{\mathrm{cell}})G_{\mathrm{federation}}J_{\mathrm{cell}}f\|^2 > 1/100.$$

Exact integer principal minors verify this for all 220 possible boundary triples. Thus its unchanged embedding has residual greater than $1/10$, independent of capacity. A dressed candidate needs a separate residual, spectral isolation, localization or propagation, and accessible-residue bound. At fixed total occupation two the raw probe variance is $8/[V(V-1)]$, so normalized spectral weight alone does not ensure feasible sampling. These are native correlation poles, with no physical particle or mass assignment (`model/native_contact_response`, [1]).

### 5.8 Connected contact response

The connected comparison has a different threshold from the isolated cell. Let $V = 240\,4^\ell$, $M = 21V/8$, and let $Q_2$ project pair coefficients onto the primitive degree-two space. The actual comparison is $H_0 = Q_2(G_2 + 2A_{\mathrm{adj}})Q_2$ on that space, where $A_{\mathrm{adj}}$ marks adjacent occupied

ports. Give every port in a cell the same auxiliary affine triangle-centroid coordinate $f$. Exact icosahedral symmetries give $\sum_i f_i = 0$, $|f_i| \le 1$, $V/5 < A = \sum_i f_i^2 \le V/3$, and $\sum_{ij\in E}(f_i - f_j)^2 < 4$. These are comparison coordinates. The primitive trial vector is

$$w_{ij} = f_i f_j - c_i - c_j, \qquad c_i = \frac{-f_i^2 + A/[2(V-1)]}{V-2}.$$

Its squared norm is at least $V^2/75$, its exchange energy at most $4V/3$, and its contact energy less than $6V$. Rayleigh's principle therefore gives

$$0 < \lambda_{\min}(H_0) \le \frac{550}{V} = \frac{55}{24\,4^{\ell}}.$$

The bound uses the exact cell adjacency and the source routing postconditions; it does not assert that the floating constructor returns at every level. The isolated pole $E > 5$ lies above this connected comparison bottom. Its isolated threshold is therefore insufficient to certify global binding. Bound bands at fixed momentum and resonances require separate tests.

The source also supplies a controlled exterior correction. Retain the 66 pair configurations wholly inside the chosen cell and write $G_2 = \left(\begin{smallmatrix} A & B^* \\ B & D \end{smallmatrix}\right)$. The literal boundary gives $A = G_{\mathrm{cell}} + N_\partial$, $B^*B = N_\partial \le 2I$ and $0 \le D \le 24I$. For $s > 0$ set

$$F(s) = A + sI - B^*(D + sI)^{-1}B, \quad F_J(s) = A + sI - \sum_{k=0}^{J} \frac{B^*(24I - D)^k B}{(24+s)^{k+1}}.$$

The core resolvent is $F(s)^{-1}$. Positive inverse ordering and the geometric remainder give

$$0 \le F^{-1} - F_J^{-1}, \qquad \|F^{-1} - F_J^{-1}\| \le \frac{2}{s^3}\left(\frac{24}{24+s}\right)^{J+1}.$$

Only finite exterior paths enter each coefficient. No projection back to the cell's primitive space is inserted during an excursion. For the normalized four-E probe $u = h/2$, $s = 24$ and $J = 2$, the connected response $s\langle u, (G_2 + sI)^{-1}u\rangle$ is enclosed between 0.660081931825 and 0.660515959604 on the evaluated cell-zero windows at levels zero through two. The isolated value exceeds the connected enclosure by more than $1/120$.

For the canonical total-two reference, this response is accessible through complete native lag experiments: with $q_s = 2M/(2M+s)$,

$$s\langle u, (G_2 + sI)^{-1}u\rangle = (1 - q_s)\sum_{n\ge 0} q_s^n C(n)/v_V, \qquad v_V = \frac{8}{V(V-1)}.$$

Truncation at $N$ costs at most $q_s^{N+1}$. Preparation errors and independent sample costs have explicit allowances; host stops remain distinct outputs. The discount is an analysis parameter. It adds no native random scheduler or physical clock. The negative-spectral-parameter resolvent bound does not continue a pole through the exterior spectrum at positive energy (`model/native_connected_contact`, [1]).

### 5.9 Positive-energy correlations from open records

On the finite configuration transfer space, the constant unit vector $\Omega$ is the unique ground of $H_M = -2M \log P_q$. It differs from the native working density $I/\binom{V}{q}$. For normalized primitive

pair fields, the complete reflected record statistics determine, using the isometric primitive two-occupation representative of $H_M$,

$$W_{ij}(t) = \langle h_i, e^{-itH_M} h_j \rangle / 4.$$

This matrix has a positive spectral measure on nonnegative energies. It is a reconstructed ground correlation, without a claimed unitary implementation on the working carrier.

The continuation has a controlled capacity dependence. On primitive degrees at most $r$, choose $b = \lfloor M/(2r\Delta) \rfloor \geq 1$, $Q = P^b \geq I/2$ and $u = 2Mt/b$. With $c_0 = 1$, $c_{k+1} = c_k(k - iu)/(k + 1)$ and $N = \max(1, \lceil |u| \rceil)$,

$$V_m(Q) = \sum_{k=0}^{m} c_k (I - Q)^k = \sum_{j=0}^{m} a_j Q^j, \quad \|V_m(Q) - Q^{iu}\| \leq 2^N - \sum_{k=0}^{m} \binom{N + k - 1}{k} 2^{-k}.$$

Substituting the actual lag-$bj$ four-E statistics gives an explicit signed complex reconstruction. Its systematic error adds $A_m \eta$, where $A_m = \sum_j (|\Re a_j| + |\Im a_j|)$ and $\eta$ bounds each normalized prepared score. For $m + 1$ settings, variance-bounded independent repetitions give sampling error at most $s$ with failure probability $p$ when each setting has at least $(m + 1)A_m^2/(pv_q^2 s^2)$ shots. This bound can be impractical; preparation, sampling and host-stop costs are not removed by analytic reconstruction. The displayed $t = 1/64$, order-12 example requires 116424316 attempts at level zero under its sufficient warmup bound. The corresponding levels one and two exceed the unchanged $2^{32}$ attempt ceiling, and the conservative 99% sampling certificate requires more than $10^{17}$ total repetitions. These are sufficient budgets, without an optimality claim.

The reference distinction has an operational witness. Bare native record effects are diagonal, so an ideal diagonal phase on the working carrier changes no complete record law, even with arbitrary initial reference correlations. In the transfer ground the same field instead has susceptibility $-2\Im W_{ii}(t)$, with a strictly positive finite-kick margin certified by native moments and a Taylor remainder. Identifying these interventions and readouts without an additional map would assign two different responses to one experiment. A physical dictionary must preserve the intervention statistics as well as the reconstructed correlations (`model/native_vacuum_response`, [1]).

**Complete finite energy windows.** The same blocked transfer defines positive effects $B_k = \binom{n}{k} Q^k (I - Q)^{n-k}$, $0 \leq k \leq n$, whose sum is the identity. Their probabilities are exact linear combinations of the complete native lag scores at $0, b, \ldots, nb$. Assign $k > 0$ the counted-energy label $-(2M/b)\log(k/n)$ and retain $k = 0$ as overflow. At a transfer eigenvalue $x \geq 1/2$, the output is binomial, so

$$\Pr(|k/n - x| \geq \delta) \leq \frac{1}{4n\delta^2}, \qquad |E_k - H(x)| \leq \frac{2M\delta}{b(1/2 - \delta)}$$

outside that exceptional event, for $0 < \delta < 1/2$. The total coefficient amplification of the measured, nonzero-lag moments is $A = 3^n - 1$. Normalized moment error $\eta$ gives histogram half-$\ell^1$ error at most $A\eta/2$. Sampling error $s$ with failure probability $p$ has the sufficient budget $nA^2/(4pv_q^2 s^2)$ independent repetitions per setting. Native local preparation on the fair 66-state source makes the $n = 16$ full-vector preparation error less than $4 \times 10^{-11}$, with at most 3871 native attempts per setting. The sufficient one-percent sampling budget exceeds $3.2 \times 10^{25}$ total repetitions. This is a reconstructed spectral statistic; the effects are not thereby an executed energy instrument. Physical mass, a laboratory energy unit and an autonomous relativistic action require separate identifications (`model/native_energy_identification`, [1]).

**A complete controlled energy instrument.** The fair 66-state carrier also admits a destructive energy measurement using its declared coherent services. An exact rational eigenframe certificate gives a nearby generator $\widetilde{H}$ with $\|H - \widetilde{H}\| \le 27 \times 10^{-8}$ in the counted convention $H = -60 \log P$, without assuming separated eigenvalues. For each certified normalized ray $u$, let $C$ be the existing source encoding and $p$ a padding address. The phase word on $z = (Cu + |p\rangle)/\sqrt{2}$ implements $U = I - 2|z\rangle\langle z|$, whose source-success block is $I - |u\rangle\langle u|$ and padding row is $-\langle u|$. Thus a padding flag detects that ray destructively; the surviving complement continues to the next filter. The sixty-six energy effects and final residual sum to the identity by telescoping. This removes the second phase word of the general weak filter, which acts trivially on this surviving complement.

All sixty-six words are separately admitted, totaling 3,853,872 pulses. Exact primitive certificates and the eigenframe residual bound give complete half-diamond error below 1/12000, including every processor failure, with stored-map gain below 1001/1000. The ideal instrument is normalized; stored outputs have that bounded gain. This is a sequential composition of declared source services, with no monolithic repair-program admission or generic-host expansion asserted. Every stored success prefix is evaluated with a retained reference control.

The record-prepared noncentral total-two reference described below feeds this same instrument, with exact acceptance 7771/65536 and explicit preparation rejection. Exact rational matrix products give the entire accepted energy law, including the residual. No successful branch is renormalized in that full output. Repair attempts and service pulses have distinct counters; physical duration and a rest-mass identification are separate maps (`model/native_energy_measurement`, [1]).

The existing source phase experiment supplies a second ideal reference $\rho = P/56 \ge I/84$ on the same configurations. Its exact modular generator is $-\log \rho = H/60 + \log(56) I$, identifying the controlled operator families under a known factor of sixty. The preparation retains its selector timeout and numerical-density admission scope; unread repair is a different evolution. This reference and the diagonal tree reference are operationally distinct: their norm-one occupation-exchange expectations differ by 1/168, so their half trace distance is at least 1/336. Their separate regional-modular and energy-clock properties cannot be combined as properties of one state.

**A fair ground, local energy and complete readout.** For the connected fair binary source put $G = \sum_e (I - S_e)$, $P = I - G/(2M)$ and $H = -2M \log P$. A triangle-cover argument gives $P \ge \alpha I$, $\alpha = 10/21$, uniformly on the growing federation. The fixed-number uniform vector $\Omega$ is the unique ground on each connected occupation sector. On the sixty-six-state source a declared coherent preparation, following eleven native tree attempts, has complete half-diamond error below 1/4,000,000. A local occupation phase creates an excitation with $\langle G \rangle = 50/33$ and $H \ge G$. The matched preparation, control and complete energy detector use sixty-eight words and 3,877,451 service pulses. Scoring all failures as zero gives an unnormalized mean-score contrast greater than 3/50; the excitation is detected without a successful-branch normalization. This is a finite excitation-energy observation, with no stable-particle or rest-mass assignment (`model/native_fair_ground_readout`, [1]).

The same source admits an explicit comparison between its logarithmic clock and local exchange dynamics. Write $\tau = I/D$, $\rho = P/\operatorname{Tr} P$, $c = \operatorname{Tr}(\tau P)$, $\mu = \operatorname{Tr}(\rho G)$ and $a = (1 - \mu/(2M))^{-1}$. For the weighted Hilbert–Schmidt norm,

$$\|[H, X] - a[G, X]\|_{2,\rho} \le \frac{\|[G, X]\| \sqrt{\operatorname{Var}_\tau(G)/c} + \frac{1}{2}\|[G, [G, X]]\|}{2M\alpha^2}.$$

Exact occupation moments bound the variance by $O(M)$ on the bounded-degree graph. Fixed local observables, times and complete classical intervention histories therefore have error $O(M^{-1/2})$. All

failures and zero slots are retained. This is a state-weighted comparison, not an arbitrary-input channel-norm claim. On $A\Omega$, positivity instead gives $\|(e^{-itH}-e^{-itG})A\Omega\| \le |t|\|[G,[G,A]]\|/(4M\alpha)$, with a corresponding fixed-word bound. A growing ground preparation is an additional premise.

Writing $h_e = I - S_e$ gives antisymmetric local currents $J_{e\to f} = i[h_e, h_f]$, supported on adjacent seams, and exact conservation of $G$. Native unread repair is a different operation: on the faithful sixty-six-state reference it increases the mean $G$ by $37/840$. For its unital channel the mean increase in $H$ is at least as large. The complete energy instrument resolves a score increase greater than 0.034 for the ideal reference, with separately counted preparation error. Source-plus-apparatus conservation requires an apparatus-energy observable. None of these graph currents supplies a symmetric Lorentz stress tensor (`model/native_local_energy_clock`, [1]).

The dilute ground imposes a separate constraint on its observable limit. For a bounded number-preserving operator $A$ on $s$ ports,

$$\mathrm{Var}_{\Omega_{V,q}}(A) \le 4\|A\|^2 \left(1 - \frac{\binom{V-s}{q}}{\binom{V}{q}}\right) \le 4\|A\|^2 \frac{sq}{V}.$$

Thus fixed-support bounded connected correlations vanish when $q/V \to 0$, uniformly in controlled time. For a fixed sequence of complete local instruments on supports $s_j$ whose fine Kraus operators each preserve local occupation, the terminal source and archive have half trace distance at most $\min\{1, 2\sum_j \sqrt{s_j q/V}\}$ from the scalar empty-sector protocol. The comparison retains every branch. A clock rescaling cannot restore a vanishing pole residue. Finite density, growing supports, rescaled fields or charged operators require separate limits; this bound does not exclude them (`model/native_dilute_ground_limit`, [1]).

### 5.10 Controlled transfer state and time

The weighted 66-state source admits a distinct operational construction using its declared coherent services. Its scalar record transfer has

$$S = \Pi^{-1/2} P \Pi^{1/2}, \quad \frac{296}{495} I \le S \le I, \quad \Omega_x = \sqrt{\pi_x}, \quad H = -\log S.$$

An eleven-E configuration detector has fine maps $\sqrt{w_h}|y_h\rangle\langle x_h|$. Retained-record routing followed by the native amplitude-tree word with rational weights $\pi_x$ changes these to $\sqrt{w_h}|\Omega\rangle\langle x_h|$. Thus the ideal prepared source factors from the retained archive and any untouched reference, even for an initially correlated input. The archive can remain correlated with that reference.

Transfer evolution has a finite residual certificate. Rational proposed columns $V$ and diagonal $(296/495)I \le \Lambda \le I$ satisfy $\|V^*V - I\| \le g = 10^{-9}$ and $\|SV - V\Lambda\| \le r = 10^{-9}$, checked by exact rational arithmetic and directed square-root intervals. The polar comparison $W = V(V^*V)^{-1/2}$ is unitary and obeys $\|W - V\| \le g$. Consequently

$$\|S - W\Lambda W^*\| \le r + 2g, \qquad \|\log S - W \log \Lambda W^*\| \le \frac{r+2g}{296/495}.$$

The logarithm bound follows by integrating the resolvent identity. The actual compiler uses normalized rational columns of $V$ and their existing rank-one phase words; it does not execute the polar map. Telescoping the 66 phases bounds their ideal unitary error by $|t|(r+2g)/(296/495) + 528g$. Scalar logarithm intervals and each stored primitive's certificate supply the additional numerical allowance.

One complete experiment prepares $\Omega$, phases configuration $a = (0, 1)$ by $i$, applies the controlled transfer evolution at $t = 1/10$, and decodes the initial configuration of a second eleven-E detector. The unphased control has probability $p = \pi_a = 625/7505$. Writing $u(t) = \langle a|e^{-itH}|a\rangle$, the intervention has

$$p_a(t) = p\,|1 + (i-1)u(t)|^2.$$

Here $H_{aa} \geq 32/495$ and $\|H\| < 103/200$. A spectral Taylor remainder proves $p_a(1/10) - p > 1/1000$. The complete protocol has 68 separately admitted coherent words, at most 3,877,451 pulses and 22 native E commands. Every first processor failure and padding abort is retained and scored zero. Passing primitive gates bound complete half-diamond error below 1/11000 per experiment and gain below 1001/1000. Charging the full trace-norm error for both rounded, potentially unequal-trace outputs leaves a stored-map score increment above 1/2000. These rounded scores are approximations to the ideal probability law.

This supplies a controlled finite preparation, intervention and readout for the source-derived transfer Hamiltonian. The nine-bit direct compiler and lazy detector retain the generic evaluator's separate resource stops. The parameter $t$ is dimensionless transfer time, not elapsed repair time or laboratory seconds. Autonomous unitary repair, a growing coherent embedding, a physical vacuum and a relativistic particle spectrum do not follow from this controlled experiment (`model/native_transfer_control`, [1]).

### 5.11 Local preparation and readout on one attempt clock

The fair federation admits a controlled preparation on each twelve-port cell, including every local occupation sector and an arbitrary untouched reference. Run its eleven-edge E tree, then $0 \leq b \leq 68$ repetitions of two ordered sweeps through the thirty internal seams. Each sixty-attempt population matrix is exactly $A_q/2^{60}$ on sector $q$. Integer enumeration for all $0 \leq q \leq 12$ proves double stochasticity and $\sum_y \min_x A_q(y,x)/2^{60} \geq 1/2$. Minorization therefore contracts half-$\ell^1$ distance by at least a factor $1/2$ per block.

This population argument has a complete quantum attachment. Each nonconstant tree record identifies a rank-one branch $2^{-11/2}|y\rangle\langle x|$; constant branches preserve the entire empty/full corner $P_c$. Retain the tree records and trace only the subsequent mixing-record values in the comparison. The resulting channel is within half-diamond distance $2^{-b}$ of the tree followed by

$$\mathcal{R}(\rho_{AR}) = P_c \rho_{AR} P_c + \sum_{q=1}^{11} \tau_q \otimes \sum_{|x|=q} \langle x|\rho_{AR}|x\rangle.$$

Here $\tau_q$ is the uniform local sector state. Its weights are inherited from the input. All physical records remain in the archive; replacing the source by $\tau_q$ while retaining every mixing record has error $1 - 1/\binom{12}{q}$ on a known basis input, regardless of mixing time. Thus this preparation does not erase archive correlations or select a global density or pure vacuum (`model/native_fair_local_preparation`, [1]).

Complete occupation readout uses the same native records. For a connected $b_0$-port window, track all incident currents, including boundary repairs. If $s_i$ is accumulated displacement, the first occurrence of an internal tree edge yields its initial contrast $d - s_a + s_b$. Tree integration fixes every nonconstant initial bit pattern. Global occupation $q$ is public source specification: empty and full windows are feasible precisely when $q \leq V - b_0$ and $q \geq b_0$. A constant pattern is therefore resolved whenever only one is feasible. In particular, for $0 < q < 12$ a fresh eleven-E intracell tree gives exact occupation readout across every feasible local sector, with its terminal disturbance and arbitrary reference.

For a general window and $0 < q < V$, $T$ uniform E attempts cover its tree except with probability $p_{\mathrm{cov}} \le (b_0 - 1)2^{-\lfloor T/M \rfloor}$. Keep coverage failures and unresolved constants as erasures. The full labelled output has half-trace distance exactly $p_{\mathrm{cov}} + (1 - p_{\mathrm{cov}})\eta_{\mathrm{amb}}$ from an initial occupation measurement followed by the same native instrument. The unresolved constant weight is $\eta_{\mathrm{amb}}$; for the canonical fixed-$q$ marginal it is bounded by $(1 - q/V)^{b_0} + (q/V)^{b_0}$ when both constants are feasible, and vanishes otherwise. This oracle is a comparison channel, not an extra implemented service. Worst-case input error need not be small (`model/native_fair_local_readout`, [1]).

Preparation and readout share the actual attempt counter. With $b = 64$, preparation followed by a fresh intracell tree takes 3862 attempts and literal nodes; the sparse-sector comparison error is at most $2^{-64}$. This two-tree program permits $b \le 67$ under the fixed node ceiling. For a 348-port certified window at level six and canonical $q = V/3$, preparation followed by $64M$ uniform attempts uses 3852 nodes and 165154571 attempts, with combined error below $10^{-16}$. These are admitted finite instruments, with expanded host stops retained. In a separate short-time comparison with binned Poisson clocks, reserving $L$ fixed native operations costs at most $KL/M$ for $K$ window seams, in addition to the existing cone, selector and joint-state errors. The long readout estimate is not identified with that short-time limit (`model/native_fair_local_clock`, [1]).

The occupation chart also binds an explicitly supplied proper-code load/export service to the actual twelve local source bits and exterior, intertwining every native C/E raw branch. After a fresh E tree, its decoded nonconstant branch selects an occupied/unoccupied seam. A beam, optional phase, inverse beam and complete compiled E detector use 305 pulses. Their ideal unconditional score increment equals the actual nonconstant branch weight $a$; certified stored primitives give score increment at least $a\,9999999/10000000$, retaining constant outcomes, all first failures and rounded number leakage. The rounded register stays in the apparatus. The supplied quantum coupling is an operational premise; pulses and native attempts have distinct counters without a derived physical time conversion. Sector-uniform mixed states alone are invariant under these number-preserving controls. No autonomous field action or particle mass follows from this conditional experiment.

**A faithful noncentral reference from the same records.** One sixty-E mixing block after the tree has exact integer transition counts $A_q[x, y]$ with denominator $2^{60}$. Every column is faithful and nonuniform on each recognized local sector $1 \le q \le 11$. Choose the explicit observer protocol $a_i = 1 + \mathrm{depth}(i)$ on the native tree and $W(x) = \prod_{i \in x} a_i$. Define $L_q = \min_{x,y} \lfloor A_q[x, y]/W(x) \rfloor$. Accepting the first $L_q W(x)$ lexicographic coin paths to endpoint $x$ prepares $\sigma_q = \mathrm{diag}(W)/Z_q$ exactly, independently of the recognized tree terminal state $y$. Exact backward path counts compute the flag from the retained tape. Conditional acceptance is at least $1/12$, and $\sigma_q \ge I/8192$. Rejects and coherent constant-code outcomes remain explicit; unconditional acceptance can vanish on a constant input. The complete source word has seventy-one attempts. The reference observation retains the tree and flag and traces only the sixty mixing values. Since $-\log \sigma_q = (\log Z_q)I - \sum_i (\log a_i) n_i$, natural number-preserving regional modular actions agree on overlaps. This observer-selected protocol supplies no unique vacuum or geometric modular clock (`model/native_noncentral_reference`, [1]).

**Occupation current and complete local histories.** For a rational profile $\phi$ on the retained ports, extend $\phi$ by zero outside and set $A_\phi = \sum_v \phi_v n_v$. Each actual seam record gives $n'_a = n_a - j$, $n'_b = n_b + j$. Thus its weighted occupation increment is $(\phi_b - \phi_a)j$, including crossing seams. A fine history is a scaled partial permutation $K_h$; if $\Delta_h$ is the summed increment, then

$$K_h^* A_\phi K_h = K_h^* K_h A_\phi + \Delta_h K_h^* K_h.$$

This proves the discrete continuity identity for arbitrary input and untouched reference. The decoded initial mask and retained currents determine every local snapshot on the same attempt clock. Keeping that initial label preserves the complete instrument comparison, so its error has no factor for the number of snapshots. Erasures receive no invented absolute trajectory. In the 3862-attempt preparation/readout join this history bound concerns the final eleven detector attempts, with preparation mixing values traced as specified. Occupation conservation supplies no physical gauge charge or stress-energy tensor by itself (`model/native_local_charge_history`, [1]).

### 5.12 Complete local records at growing capacity

The capacity comparison also admits complete local quantum instruments. Fix observed seams and final ports with endpoints in a finite set $S$, and a radius-$r$ neighbourhood $B$ with $v$ ports and $K$ internal seams. Retain every observed native tag, coin, raw value and current, its order, and the final quantum state with an untouched reference. Remote records and exact global tick addresses are traced by this observation, while remaining available in the native archive. Choose a common partition $0 = t_0 < \cdots < t_b = t$ with $Mt_j$ integral and run $T = Mt$ native attempts. Each observed event is assigned its partition bin; its order within the bin is retained. Here $t$ is per-seam attempt time, twice the exchange time above, with no physical calibration.

The complete instrument differs from its restriction to $B$ by at most

$$a_N = \min\left\{1, |S|\binom{T}{r+1}(\Delta/M)^{r+1}\right\}$$

in half-diamond norm. Conditional on the input-independent selector word, the backward cone seeded at terminal $S$ contains every observed record root. Leaving $B$ requires a time-ordered path of $r+1$ seams, which proves the bound for arbitrary coherent inputs and references.

Compare the restricted process to independent rate-one Poisson attempt clocks on its $K$ seams, applying the same native instrument at each event. In one interval of length $1/M$, the native window selector is empty with probability $1-\lambda$, or has each seam with probability $1/M$, where $\lambda = K/M$. The corresponding Poisson probabilities are $e^{-\lambda}$ and $e^{-\lambda}/M$, with every multiple-event word retained. Their exact total variation is $\lambda(1-e^{-\lambda}) \le \lambda^2$. Coupling all $T$ intervals and applying the identical conditional quantum maps therefore gives

$$\tfrac{1}{2}\|\Phi_B - \Phi_{\mathrm{Pois},B}\|_\diamond \le d_N := \min\{1, tK^2/M\}.$$

Aligned boundaries give the same bin label on matched events. Exact continuous timestamps have a singular comparison with native grid times, and the full global record-count laws differ, so neither observation can replace the declared local binned one.

If initial joint window/reference states differ by half-trace distance $\epsilon$, their complete observed outputs differ by at most $\min\{1, \epsilon + a_N + d_N\}$. For a canonical fixed-$q$ source and an independent common reference, comparison to a declared Bernoulli density $p$ gives

$$\epsilon \le \min\{1, v(v-1)/(2V) + v|q/V - p|\}.$$

Matching rooted windows thus give an explicit comparison between source capacities. If all fixed-radius windows stabilize and their state restrictions converge, the additional Poisson cone bound $|S|(\Delta t)^{r+1}/(r+1)!$ constructs a limiting local instrument. The evaluated source levels zero through two have identical cell-zero radius-one windows with 15 ports and 33 seams, with a verified orientation-preserving record map. This establishes neither a unique all-radius routed refinement nor the physical density or spatial dimension. The proof and exact rational controls retain every overflow and distinguish kernel analysis from bounded native execution (`model/native_local_history_limit`, [1]).

### 5.13 Rooted source refinement

The exact oriented midpoint stencil supplies a constructive rooted cell topology. Choose the cell that follows child zero at every subdivision around one original vertex. Inserting an additional leading child zero identifies its radius-$r$ dual ball, including incident edges, whenever $2^\ell > r + 2$. In each of the five incident original faces, a dual step increases the triangular grid's radial coordinate by at most one. This proves stabilization of every finite corner ball and consistency under restriction. The resulting countable graph retains the five-face corner; no assertion about an arbitrary root sequence follows.

A recursive local evaluator also computes the literal midpoint coordinates, face orientation and port scores without allocating the sphere. It reproduces coordinates bit for bit and routed neighbor rows at the independently evaluated levels zero through three. Exact binary32 score intervals certify first-maximum decisions, including stable ties. Collision repairs require a unique certified assignment. Uncertain decisions are rejected. The finite address domain extends through level fourteen, where every source edge key fits its signed integer representation. These are finite numerical attachments to the oriented topology.

The certified tables give matching windows of cell radii one, two, four and eight at specified levels through fourteen. Keeping every port of a radius-$r$ cell ball contains the required radius-$r$ port ball. Canonical names preserve source endpoint order and therefore intertwine the complete native instruments, including signed currents and zero slots. For declared density $p$, the canonical totals $q_\ell = \lfloor p\, 240\, 4^\ell \rfloor$ have the local state bound above; the Bernoulli comparison states restrict consistently. The radius-four, level-fourteen window has 348 ports. The observation retains records on the thirty root-internal seams and the final state on the twelve root ports; other window records are traced. At $p = 1/3$ and $t = 1/630$, its complete local-output comparison bound is below $10^{-6}$. The corresponding 268435456-attempt description is admitted. The expanded host retains its unevaluated-carrier stop; this is a finite kernel bound, without a claimed execution of the global experiment.

An infinite routed process requires a specified arithmetic continuation. Finite-width level tags and coordinate collapse in bounded local tests prevent treating the literal numerical constructor as a total such specification. Exact coordinates, arbitrary precision and combinatorial port assignment are different continuations. The construction chooses none of them and establishes no unique all-radius routed limit, physical density or Lorentzian field (`model/native_rooted_refinement`, [1]).

### 5.14 Native clock observables

The declared coherent clock services also give operational noncommuting measurements on the same biased 66-state source. On the existing seam $(1, 7)$, pair $|1, k\rangle$ with $|7, k\rangle$ for each of the ten spectators. Let $X, Y, Z$ be the direct sum of the usual Pauli matrices on these pairs, zero on the 46-dimensional complement $P_0 = I - Q$. The existing balanced cap and quadrature words give analyzers $U_A$ with $U_A^\dagger Z U_A = A$. One native contrast-reading repair after the analyzer therefore has effects

$$P_{A,\pm} = \frac{Q \pm A}{2}, \qquad P_{A,0} = P_0, \qquad A \in \{X, Y, Z\}.$$

All 25 raw addresses are retained, including 19 zero maps. The six live maps have coin factors $8/33$ and $25/33$. The zero-contrast maps preserve all coherences within $P_0$, including coherence between empty and doubly occupied endpoints. The raw update is destructive. Its retained displacement record permits an explicit correction: apply two existing balanced cap rotations when the repair moved, then undo the analyzer. Every fine corrected Kraus operator is a scalar sign times $\sqrt{p_{\mathrm{coin}}} P_{A,\delta}$

on all 66 states. This gives the three-outcome Lüders instrument, with its coin refinement and arbitrary untouched references.

For the full normalized stationary density, no odd-sector conditioning is needed:

$$\langle X\rangle = \langle Y\rangle = 0, \quad \langle Z\rangle = \frac{1649}{7505}, \quad \operatorname{Var} X = \operatorname{Var} Y = \frac{3201}{7505}, \quad \frac{|\langle [X,Y]\rangle|^2}{4} = \left(\frac{1649}{7505}\right)^2.$$

These actual observables and state supply the finite Robertson theorem's inputs. With $\Delta = \log(25/8)$, the working modular clock acts by

$$U_s = \rho_*^{-is}, \qquad U_s^\dagger X U_s = \cos(s\Delta)X - \sin(s\Delta)Y.$$

The literal nine-qubit analyzer words have full-address certificates, retained export failures and a separate multiplication-rounding allowance. These bounds concern the stored numerical primitive matrices, with no hardware or unrounded analytic exponential error claim. The coherent services and their feedback composition are declared observer operations; they are outside the bare repair/hold grammar. This finite measurement menu supplies neither universal implementation independence for every complex effect nor a physical position–momentum relation (`model/native_clock_observables`, [1]).

**Configuration readout and repeatable preparation.** There is a bounded contrast-record detector on this same sector. Perform one native E repair on each edge of a fixed spanning tree of the twelve ports. Let $u_i$ be the accumulated recorded change of occupation at port $i$. An incoming contrast $d$ on $(a, b)$ gives the initial difference $n_a(0)-n_b(0) = d-u_a+u_b$; the retained current updates $u$ after that read. The eleven differences and $\sum_i n_i(0) = 2$ determine the initial configuration. Thus every nonzero fine history has a rank-one map $K_h = \sqrt{p_h}\,|x_h\rangle\langle x_0|$, and coarse graining by the decoded $x_0$ gives the full computational projective measurement. All $25^{11}$ formal raw words are retained in an exact lazy representation; $66\,2^{11} = 135168$ words have nonzero maps. One queue slot and ten exports suffice for the eleven-step record protocol. Streamed branch continuation uses the existing native transition kernel and authenticates its records. The stock expanded evaluator's resource stop after three steps is retained as a separate return; it is not equated with the lazy evaluator. The nine-bit coherent words use the direct source compiler; the generic word evaluator has a six-qubit host ceiling.

The terminal configuration is also known from the currents. Existing coherent bit routing maps its code either back to $x_0$ or to a fixed target code. The ideal corrected maps therefore give the basis Lüders instrument or the replacement channel $\rho \mapsto |x_*\rangle\langle x_*|\operatorname{Tr}\rho$, including untouched references. This supplies repeatable preparation to the existing pair-analyzer architecture: its 4291 settings span all 4356 Hermitian coordinates. Complete compiler bounds and export outcomes accompany these compositions. The rounded maps have bounded normalization error; their positive record functionals retain full rank without output renormalization. Coherent service semantics are declared, and finite tomographic coverage does not prove universal coexistent-effect realization or a growing field-observable family (`model/native_effect_coverage`, [1]).

**Rank-one measurement synthesis.** The supplied coherent services implement more than the fixed tomographic menu. For any nonzero Gaussian-rational vector $v$ on the 66 source states and rational $0 \le \lambda \le 1$, with each input numerator and denominator of at most 64 bits, consider $E = \lambda|v\rangle\langle v|/\langle v, v\rangle$. In the existing 512-address register, append the amplitude $\sqrt{1-\lambda}$ at one padding address. A binary amplitude tree and diagonal phase word prepare the resulting normalized vector from address zero. Executing the inverse word and routing zero to a genuine source code

before complete export and tree readout gives $E$ as the ideal selected effect. All other source outcomes, padding aborts and processor failures are retained. No extra register or conditioning on source-code success is used, and an effect alone specifies no Lüders update.

Directed integer trigonometric enclosures select a 41-bit angle grid; exact Walsh identities compile its layers into existing native pulses. The full-register operator error from target angles is at most $11\,2^{-41}$. The emitted word uses at most 23579 pulses, below the unchanged source-word limit. The certificate checks every emitted stored six-dimensional primitive, including negative inverse requests and fixed basis-change angles, against its full ideal matrix and rejects a failed bound. For a passing word with $N$ pulses and $g = 2\,10^{-11}$, put

$$x = Ng, \qquad e = \frac{x}{1-x} + 11\,2^{-41}, \qquad F_N = \frac{Ng^2}{(1-x)^2}.$$

Telescoping the good-code products and retaining every first processor failure gives complete half-diamond error at most $[e(2+e) + F_N]/2 < 10^{-6}$, including arbitrary untouched references. These are bounds on the stored primitive maps; floating propagation arrays are diagnostics. The direct nine-bit compiler executes the tape, while the generic six-bit word evaluator returns its resource stop. The real projector onto $(3|0\rangle + 4|65\rangle)/5$ also has an 80-pulse construction with error below $2\,10^{-9}$, despite lying outside the positive cone of the fixed balanced-analyzer menu. Fixed precision and word length supply neither arbitrary-accuracy dense effects nor universal coexistent refinement. The composition consumes the declared coherent services and Kraus probability semantics; it does not derive those premises from repair counting (`model/native_effect_synthesis`, [1]).

**Complete coexistent weak measurements.** The same source services give joint noncommuting effects with their disturbance. For a normalized rational-direction vector $u$, projector $P = |u\rangle\langle u|$ and rational $0 \le s \le 1$, couple $u$ to one existing padding address by a rank-one phase. Complete source export and a second correcting phase give the ideal surviving contraction $L = I + (s-1)P$ and abort effect $(1-s^2)P$. The abort retains its reference state and terminates; the success remains a quantum input to the next stage. For at most eight such stages, put $A_0 = I$ and $A_j = L_j A_{j-1}$. The complete effects are

$$E_j = A_{j-1}^*(1-s_j^2)P_j A_{j-1}, \qquad E_{m+1} = A_m^* A_m, \qquad \sum_{j=1}^{m+1} E_j = I.$$

The last identity telescopes, including every earlier disturbance. With $u_1 = |0\rangle$, $s_1 = 3/5$, $u_2 = (|0\rangle + |65\rangle)/\sqrt{2}$ and $s_2 = 4/5$, the $(0, 65)$ entry of $[E_1, E_2]$ is $216/3125$. Their coexistence follows from one common complete instrument.

Each stage uses two separately admitted 58392-pulse words in the existing nine-bit work register. Every first processor failure, padding abort and residual outcome is retained, with no intermediate destructive tree readout. Conditional on all exact primitive checks passing, a gain-aware complete half-diamond bound for eight stages is below $1/50000$. The generic evaluator retains its resource stop; the direct compiler and lazy complete instrument have their own stated domain. This bounded family supplies noncommuting coexistent effects but does not implement every effect or discharge the universal operational Born premises (`model/native_coexistent_effects`, [1]).

**A prepared phase contrast under fair repair.** A separate twelve-bit occupation-register experiment targets the fair source's mixed state $\rho = P/\operatorname{Tr} P$, where $P = \frac{1}{30}\sum_e Q_e$, $Q_e = (I + \mathrm{swap}_e)/2$ and $\operatorname{Tr} P = 56$. Declared coherent services target the ideal successful blocks $q\rho$ and $qD\rho D$,

with $D = (-1)^{n_0}$. Their ideal signed single-occupation exchange readouts on ports $(0,1)$ are $\pm q/168$, $\pm 5q/1008$ and $\pm q/240$ before repair, after C and after E, respectively, where $q = 1 - (368/2048)^{16}$ is the unconditional selector acceptance. The exact difference operator is multiplied by $5/6$ under C and $7/10$ under E, although the two repairs share their population transfer. A separate rational implementation checks the numerical source execution's prepared density blocks with Frobenius tolerance $2\,10^{-10}$, checks outside-carrier mass separately, and compares every outcome with its target. Source failures, selector timeout and all terminal labels are retained. These bounded numerical checks supply no arbitrary-reference channel-error certificate. The mixed preparation differs from the transfer ground; its decaying phase contrast identifies neither a protected sector nor a particle mass. Matching a population transfer alone does not establish a common state-and-instrument dictionary (`model/native_phase_experiment`, [1]).

### 5.15 Protected charge and its native reference

An explicit composition of the biased native repair and its persistent ordering bit connects protection to a state and a modular clock. For the binary total-two source, let $\theta_0 = \theta_1 = 25$ and $\theta_i = 8$ otherwise. Its reference is $\rho_* = 7505^{-1} \sum_x \prod_i \theta_i^{x_i} |x\rangle\langle x|$. The ordered branch is a weighted partial injection with an ordering flip $X$ when transport crosses the occupied spectator. All branches commute with $Q = I_{66} \otimes X$. Their complete star algebra has commutant $\mathrm{span}\{I, Q\}$. A faithful invariant state makes this the adjoint fixed space, so the stationary densities are exactly

$$\rho_\alpha = \rho_* \otimes [\alpha P_+ + (1-\alpha) P_-], \qquad P_\pm = (I \pm X)/2, \quad 0 \le \alpha \le 1.$$

The weight $\alpha$ is conserved. With the ordering bit prepared in either sharp sector, 1600 unread attempts prepare the source/reference marginal within $10^{-6}$ of its stationary product; the complete archive is not claimed independent. The proof includes arbitrary untouched references.

For $0 < \alpha < 1$, put $K_\alpha = -\log \rho_\alpha$. A recorded current $j$ on seam $(a,b)$ obeys $[K_\alpha, K_h] = j \log(\theta_a/\theta_b) K_h$. Thus every complete branch is covariant under the same modular clock. The negative exchange sector has the actual two-occupation CAR transport sign. Its closed odd exchange returns to the initial occupation and has zero modular energy change. The two protected sectors differ in mean modular energy by $\log[\alpha/(1-\alpha)]$, set by their conserved reference weight. Protection therefore supplies no nonzero rest energy. The potential commutes with occupations; its unitary flow is not the stochastic motion. These statements concern the composed 132-state source, whose ordering memory is an additional declared service (`model/protected_reference`, [1]).

The fixed-selector raw instrument has the exact weighted reverse $\rho_\alpha^{1/2} K_h^\dagger \rho_\alpha^{-1/2} = K_{\bar{h}}$, where a label involution exchanges each moving outcome with its reverse and fixes the holding and zero slots. This proves Kubo–Martin–Schwinger detailed balance and equality of stationary fixed-length record probabilities under reversed order and labels. At $\alpha = 0, 1$ the identity uses the invariant supported corner. It concerns stationary preparation and this selector; it supplies neither a reversal theorem for arbitrary feedback nor physical time reversal (`model/protected_reference/REVERSAL.md`, [19, 1]).

**Complete gauge transport.** The same literal quantum-link matrices supply a complete locally covariant operation for each of the five matter blocks, including colored matter. For a block of dimension $d$, let $A = \sum_{ab} |a\rangle\langle b| \otimes U_{ab}$ and $B = A/\sqrt{d}$. Both the raising coordinates and their conditional rotor completion obey $(A^\dagger A)^2 = d A^\dagger A$. On the two-position edge define

$$T = |x\rangle\langle y| \otimes B, \qquad P_y = T^\dagger T, \quad P_x = T T^\dagger.$$

The endpoint intertwining equations imply that $T, P_x, P_y$ commute with the total local Gauss action. Since $P_x P_y = 0$, the complete directional instrument is $\{T, I - P_y\}$, its reverse is $\{T^\dagger, I - P_x\}$, and $W = T + T^\dagger + I - P_x - P_y$ is a Hermitian unitary. Every blocked flux state is retained. The exact identities cover all sixty link states and all twelve currents; the hard-core spectator contributes the usual CAR sign.

The covariant unitaries $U(c, s) = I + (c - 1)(P_x + P_y) - is(T + T^\dagger)$, $c^2 + s^2 = 1$, expose the unselected operation parameter. On a fixed gauge-singlet preparation with a dual charge reference at $y$, the complete detector $\{P_x, I - P_x\}$ gives probability $s^2$. The rational choices $(c, s) = (3/5, 4/5)$ and $(5/13, 12/13)$ give $16/25$ and $144/169$ on that same experiment. Covariance and completeness therefore do not determine its evolution. This finite operation uses a declared matter/link extension, without equivalence to the original C/E law or a full collision extension (`model/gauge_defect_transport`, [1]).

The protected reference and this transport admit a specified common edge process. On native seam $(0, 11)$ the move $\{5, 11\} \to \{0, 5\}$ crosses the spectator, so its joint operator is $\widetilde{T} = |\{0, 5\}\rangle\langle\{5, 11\}| \otimes X \otimes B$. The forward and reverse coins have probabilities $25/33$ and $8/33$. Their complete transport/hold instruments preserve $Q$ and the reference $\rho_* \otimes \sigma_\alpha \otimes I_{60d}/(60d)$, where $\sigma_\alpha = \alpha P_+ + (1 - \alpha) P_-$, by exact endpoint detailed balance. They retain the other 64 configurations as holds. This full reference is gauge invariant without support on the joint Gauss kernel. A declared dual reference gives an orthonormal Gauss-singlet pair $\psi_x, \psi_y$ and stationary corner state $(25|\psi_x\rangle\langle\psi_x| + 8|\psi_y\rangle\langle\psi_y|)/33$. In a fixed memory-charge sector one unread attempt replaces every admitted corner input by this mixture, including reference-entangled inputs. Its complete records are retained before taking that marginal. The same modular reference assigns forward energy change $\log(8/25)$. This is a declared one-edge law, with a designated moving internal fiber and separate singlet preparation. It supplies no all-seam gauge network, global mixing theorem, two-particle colored identity or selected coherent Hamiltonian (`model/protected_reference/GAUGE_JOIN.md`, [1]).

The same corner supplies a finite thermodynamic instance. Summing all four outcomes gives the diagonal kernel $P(i, j) = p_j$, $p = (25/33, 8/33)$. With $E_i = -\log p_i$ and multiplier one, the Lean theorem `corner_same_kernel_four_laws` identifies this kernel with the heat bath consumed by the finite four-law core (`Lean/Thermodynamics/ProtectedGaugeCorner.lean`, [1]). It gives the first-law split, relative-entropy contraction and multiplier identifiability. The energy is nonconstant, so the replacement exchanges modular heat for some inputs. Its fixed-spectrum mathematical Gibbs family has zero low-temperature entropy; the fixed native coin supplies no implemented temperature variation, physical calibration or uniform capacity limit.

**Gauge-invariant propagation on trees.** The configured sixty-state quantum-link Hamiltonian has a complete spatial sector on a finite rooted tree $\mathcal{T}$. Retain all five native matter blocks, fix total fermion number one, and put a static dual reference for one block $r$ of dimension $d_r$ at the root. Every block has nonzero central charge and scalar current commutant. Leaf elimination therefore forces all links off the root-to-fermion path to vacuum; irreducibility and inequivalent central charges leave one contraction along that path. The entire Gauss kernel has an orthonormal basis $|v; r\rangle$, one state for every vertex, including the root.

For the owning literal raising-link hopping of strength $J$ and the electric occupation-projector cost $\epsilon \geq 0$, the exact restriction is

$$H\big|_{\ker G} = J\sqrt{d_r}\ \mathrm{Adj}(\mathcal{T}) + \epsilon\, \mathrm{diag}(\mathrm{dist}(0, v)).$$

The factor $\sqrt{d_r}$ comes from the norm of the contracted flux string; substituting normalized partial-isometry hopping would change the configured action. Position projectors commute with Gauss

and form a complete detector, with every position and repeated-position outcome retained. Their readout can change hopping energy, whose exact work balance is accounted for.

On an additionally specified infinite bounded-degree tree, positive $\epsilon$ gives a self-adjoint operator with compact resolvent: its diagonal potential tends to infinity and hopping is bounded. For initial support inside radius $R_0$, compression to radius $R$ gives the complete continuous-time position-record total-variation bound

$$\min\left\{1,\ 2\sum_{n\geq R-R_0+1}\frac{(D_{\max}|J|\sqrt{d_r}\,t)^n}{n!}\right\},$$

where $D_{\max}$ bounds degree and $t$ is the total evolution time across all readouts. The executed Cayley clock has a separate finite-step comparison allowance. These reference-bound dressed levels use a specified graph, link, coupling and clock. They supply no freely propagating asymptotic particles, relativistic dispersion or physical mass (`model/native_gauge_strings`, [1]).

**Source coupling readout and its color-action boundary.** On one link, the same Gauss-singlet source states give the exact block

$$H_r=\begin{pmatrix}0 & J\sqrt{d_r}\\ J\sqrt{d_r} & \epsilon\end{pmatrix},\qquad p_1(t)=\frac{4d_rJ^2}{\epsilon^2+4d_rJ^2}\sin^2\left(\frac{t\sqrt{\epsilon^2+4d_rJ^2}}{2}\right).$$

Here $p_1$ is an unconditional endpoint probability; the other outcome and all measurement back-action are retained. For $J\neq 0$, $p_1''(0)=2d_rJ^2$ and $\epsilon^2=-p_1''''(0)/p_1''(0)-2p_1''(0)$. Two finite-time probability intervals at $t$ and $2t$ supply rational Taylor enclosures, avoiding an assumed derivative oracle. The configured source sets $J=\epsilon=1$ in its model clock, hence determines the clock-independent ratio $\epsilon/|J|=1$. Reading these specifications is sufficient to fix that model ratio.

The admitted color currents have $\mathrm{tr}_3(T_aT_b)=\delta_{ab}/2$, whereas their selected fifteen-state matter lift has $\mathrm{Tr}_{15}(T_aT_b)=2\delta_{ab}$. These trace conventions must not be interchanged. The five nonvacuum source link sectors have $(C_3,C_2,Y^2)$ equal to

$$(0,0,1),\ (4/3,0,1/9),\ (4/3,3/4,1/36),\\ (0,3/4,1/4),\ (4/3,0,4/9).$$

Their configured electric energies are all one. No coefficients $a,b,c$ give this vector as $aC_3+bC_2+cY^2$: the second and fifth entries force $c=0$, contradicting the first. The zero-energy link vacuum excludes an added common identity. Thus the source electric law cannot be directly identified with a quadratic gauge Laplacian. This does not exclude an emergent color action after a controlled coarse-graining.

For a separately justified continuum action $S_E=(\kappa/2)\int \mathrm{tr}_3\,\mathcal{F}_{\mu\nu}\mathcal{F}_{\mu\nu}$, with $\mathcal{A}=g_sA$, $\alpha_s=1/(4\pi\kappa)$ in the specified normalization. The finite gauge-invariance theorem permits every positive $\kappa$. The missing source identification is the continuum action coefficient and its renormalization scheme and scale. Running a supplied coupling does not select this boundary value. The exact finite readout and coefficient comparison are in `model/source_strong_coupling` [1].

**Complete color-loop readout in the interacting Higgs source.** The separate Casimir-weighted action retains its full 438- and 1,596-state Gauss sectors, with empty colored-fermion occupation, three identity family matrices and the stated scalar cutoff. Write $H=eE-m(L+L^*)+hT+gS+sN_H$, where $S$ is the actual Higgs vertex and $T$ includes the hopping sign. Its three

colored link blocks have $(d_r, c_r) = (3,4), (6,25/8), (3,3)$, with $c_r = 2C_3 + C_2/2 + (6Y)^2/12$. The vacuum input and its actual reference-neutral addition image are labeled $q = 0, 1$. The complete instrument resolves scalar occupation and both link blocks; all output states and zero-probability records are retained. Summing both scalar occupations in the equal-block loop outcome gives

$$\frac{p_r(t)}{d_r^2} = m^2 t^2 - m^2 \left( \frac{e^2 c_r^2}{3} + 18m^2 + \frac{qh^2}{3} \right) t^4 + O(t^6).$$

The exact full-source moments cancel the Higgs contribution at fourth order; a sixth-order control detects its nonzero interacting contribution. For the two dimension-three blocks, let $M$ be the quadratic coefficient divided by nine and $D$ the difference of quartic coefficients divided by nine. Then $M = m^2$, $D = -7e^2m^2/3$ and $(e/|m|)^2 = -3D/(7M^2)$. This reads a clock-invariant bare ratio when $m \neq 0, e > 0$. Six positive times $t_j = j/1024$, exact polynomial evolution through order 28 and norm-controlled remainders enclose the default value one. The full source, diagonal energies and detector labels are regenerated. The preparation, flux instrument and Hamiltonian clock are declared finite interfaces. No repair-only flux compiler, chiral colored continuum or renormalized strong coupling is inferred (`model/color_loop_readout`, [1]).

**A local symmetry-sector excitation.** On the actual open eight-site selected spin ground at $J = 1, B = 1/4$, a one-site $Z$ pulse flips the global internal $X$ parity and leaves every initial exterior expectation unchanged. Second quantization includes the minus sign on doubly occupied sites; the full centered Hamiltonian and every complete recorded C/E branch commute with this parity. The protected single-occupancy spin code has identity repair branches with fair coin records. The ideal selected-state energy excess is enclosed by $[1.825861195059, 1.825861195062]$. Complete native preparation and detection execute in 1,090 and 1,092 pulses and certify pointer contrast above 0.67. The local commutator bound and complete source/spin comparison retain contrast above $2/5$ through at most ten repairs and time $1/16$ on $c = 12^4, U = 12^8$. This lifetime tape is an analytic construction; evolution spreads the excitation. The parity is a finite symmetry charge, not a topological boundary invariant or a physical mass (`model/defect_localization_readout`, [1]).

**Continuous charge, creation work and conjugate fields.** The exact full-Fock spin generator induces $Q = \sum_x \epsilon_x X_x/2$ in the Marshall frame. It commutes with the source action and every recorded repair branch. Ground uniqueness and a charge-reversing source automorphism force $Q\Omega = 0$. The complete local instrument $A_q = (Z_0 - qi\epsilon_0 Y_0)/2$, $q = \pm 1$, obeys $[Q, A_q] = qA_q$ and $\sum_q A_q^* A_q = I$. A local $Z$ followed by one $X$ pointer prepares these coherent branches. Measuring every local $X$ and grouping the outcomes gives the charge probabilities but dephases within a sector, so it is a different preparation.

The original weights for $q = -1, +1$ are approximately 0.83547507093 and 0.16452492907. A complete native Pauli estimator and an independent controlled-clock phase measurement recover the unconditioned work $W_q = \langle A_q^*[H_B, A_q] \rangle$. The conditional mean excitation energies are approximately 1.84463098607 and 1.73054620199 at $J = 1$. Both channels have positive spectral variance. Their first return-survival value 0.99 occurs in the certified source-time intervals $[0.1726210202, 0.1726210331]$ and $[0.1044566248, 0.1044566376]$, respectively. Return survival can rebound. A fixed uniformly chosen local-$X$ detector retains original-weight charge contrast $w_q/8 > 1/50$ at every source time. Thus charge conservation, return retention and a particle decay lifetime are distinct assertions. The generator action over an autonomous interval is $TE_{\mathrm{exc}}$ while ongoing system power is zero. Scalar bare-repair effects cannot measure these different energies; the coherent controls supply the sensitive readout (`model/source_energy_effort` and `model/charge_channel_lifetime`, [1]).

For the ordered source, define the second staggered configuration

$$Q_{2,p}(F) = p^2 \sum_x \epsilon_x F(px) S_x^2.$$

Append its bounded groups to the existing $Q_{3,p}, P_{2,p}$ histories. Internal rotation symmetry supplies its variance and energy bounds. The equal-time estimate

$$\|[Q_{2,p}(F), Q_{3,p}(G)]\| \le \frac{p^4}{2} \sum_x |F(px)G(px)| = O(p)$$

and the full-history transport bounds construct commuting configuration groups on a common further subsequence. The source U(1) acts there with charged vectors $\Phi_\pm \Omega = (Q_3 \mp i Q_2)\Omega$. An odd microscopic translation combined with the internal axis-two $\pi$ rotation induces charge conjugation, commuting with the limiting energy and spatial translations. It gives

$$\mu_{\Phi_+\Omega} = \mu_{\Phi_-\Omega} = 2\mu_{Q_3\Omega}.$$

These measures have nonzero positive gapless spectral weight. No second uniform canonical momentum is appended, and no Lorentz spin, sharp shell or physical antiparticle name follows from this internal symmetry (`model/protected_defect_catalog`, [1]).

For a normalized source-created vector $a \in \operatorname{Dom} K$, retain its full complement: $r = (I - |a\rangle\langle a|)Ka$ and $D = K$ compressed to that complement. Then

$$\langle a, (K+z)^{-1} a\rangle = \left[z + \langle K\rangle_a - \langle r, (D+z)^{-1} r\rangle\right]^{-1}.$$

The positive memory term is nonzero in both finite charge channels. Bounded sine preparation and the source resolvent filter construct this response on the same continuum history subsequence with nonvanishing original probability. Their filtered first two moments converge through bounded spectral integrands. This retains the spectral complement without fitting a mass or assuming a one-frequency field (`model/spin_dynamical_memory`, [1]).

**Bounded charge and pair preparation.** The same infrared integral, evaluated with exact rational upper bounds, and the biased trial $(I + tM)G$ with $t = 10$ certify the selected order $m > 1/10$. For a unit test in $r/2 \le |k| \le r$, write $x = \sqrt{r}\, Q_3(F)$ and $y = \sqrt{r}\, Q_2(F)$. Define

$$B_+ = \frac{1}{2\pi} \int_0^{2\pi} e^{-i\theta} \frac{x\cos\theta + y \sin\theta}{1 + (x\cos\theta + y\sin\theta)^2}\, d\theta, \qquad B_- = B_+^*.$$

Each norm is at most $1/2$, so these operators and $(I - B_+^* B_+ - B_-^* B_-)^{1/2}$ form a complete instrument. The canonical commutator, configuration variance, and first/inverse energy moments give original joint probability at least $p_* = 1/1757600$ for each charge and the band

$$r/2 \le |P| \le r, \qquad 3p_* r \le L \le r/p_*.$$

All complementary events are retained. This uniform lower bound concerns a broad band and gives no spectral pole. Bounded source quadratures converge on the same ordered history subsequence (`model/spin_charge_dispersion`, [1]).

Since translated configuration fields commute, finite spatial averaging of $A = B_+^* B_+$ gives arbitrarily far rational displacements with $\|\alpha_a(B_+) B_+ \Omega\|^2 \ge p_*^2/4$. This uses positivity and translation invariance without clustering. For every fixed $n$, Jensen's inequality applied to the spatial average

of $A$ gives separated preparations of every charge-sign pattern, with original weight at least $p_*^n/4$ and all $3^n$ records retained. An explicit cube contains suitable profile centres; selecting them effectively also requires the source-state readout bounds. The corresponding charge-$-2$ statement follows by conjugation. All nine joint preparation outcomes remain. The source energy-form bound gives $\langle L\rangle_{\text{unnormalized}} \le 27r/4$ for each pair; an upper cutoff $54r/p_*^2$ retains at least $p_*^2/8$ original probability, possibly including zero energy. On the finite cube the pair columns span each entire 28-dimensional charge-$\pm 2$ sector, with a complete detector and connected retarded response. Asymptotic factorization, a comparison dynamics and a controlled long-time limit are additional requirements for particle scattering (`model/source_charge_scattering`, [1]).

**Local creation, positive transfer and the scattering comparison.** The actual canonical translations shift the first configuration while fixing the second. Their strong continuity, the configuration rotation symmetry and Fubini's theorem imply absolute continuity of every finite independent joint configuration law. No second canonical momentum is assumed. Nonzero polynomials in distinct bounded charge creators have strictly positive vacuum norm; their finite Gram matrices are nondegenerate. This does not supply a tensor-product particle inner product.

Cutting off an annular test in physical space gives compactly supported bounded charge creators. A direct canonical-commutator estimate retains original joint-band probability $9/200000000$ for each charge, with $r/2 \le |P| \le r$ and $ar \le L \le br$, where $d = 3m_0/1000$, $a = 3d^2/2$, $b = 2/d^2$ and $m_0 = 1/10$. Time filtering with the same source clock gives $C_+, C_-$ of norm at most $1/2$ and positive energy transfer in $[ar/2, 2br]$. Analytic Fourier norm bounds fix a scale-independent normalization $M$; the inner-band probability is at least $p = 9/(200000000M^2)$. The positive square-root complement completes the three-record instrument. Both operators raise energy and are exchanged by charge conjugation; they are not each other's adjoints.

The inherited causal cone bounds their separated commutators and time derivatives. Applying Jensen's inequality to the spatially averaged positive effect, and controlling the noncommuting product by those bounds, gives equal-sign $n$-charge preparations of original probability at least $p^n/4$. All $3^n$ records remain. The selected n-charge branch has its entire energy in $[nar/2, 2nbr]$; the complementary records need not lie in that interval. Every nonzero integer internal charge therefore has positive spectrum approaching zero. This statement does not identify each charge with an elementary massless species.

For a proposed comparison dispersion $w(k)$, the actual single-packet derivative norm squared is

$$\int |E - w(k)|^2 |\widehat{f}(k)|^2 \, d\mu_{C\Omega}(E,k).$$

It is independent of time. A fixed broad energy band alone cannot make this defect integrable on an infinite time tail. The exact product derivative estimate separates this single-packet term from the controlled inter-packet commutators. Minimizing over measurable $w$ leaves the conditional energy variance at fixed momentum. Narrowing a filter onto a zero-weight graph loses its original success probability; a preparation with positive variance can nevertheless contain a distinct sharp subbranch. Source spectral branches, comparison-space normalization and long-time estimates are additional scattering inputs (`model/spin_multipacket_comparison`, [1]).

**Spatial quantum numbers and a common matter clock.** The actual proper cubic site rotations preserve the selected source and act on its charged fields by scalar test-function pullback. The binary-cubic central two-pi element consequently acts trivially. This does not determine a continuum spin: the harmonic polynomial $x_1^4 + x_2^4 + x_3^4 - 3|x|^4/5$ is a cubic singlet in spin four. The local spin creators are even full-source fermion bilinears and commute on disjoint sites. A physical

spin-statistics assignment requires their continuous spacetime representation and spectral-sector attachment.

The full-ground protected projection also constrains a common matter limit. With $e$ graph edges, $U \geq 4ce$, $R = U/c$ and exchange $J_{\text{eff}} = c^2/U$, the occupation-charge gap divided by the ballistic spin-clock scale is at least

$$\frac{R(R/2 - e)}{p} \geq \frac{4e^2}{p}.$$

It diverges under spatial refinement. Original odd fermion creation weight in bounded-energy experiments therefore vanishes on this clock. A chemical reference shifts conjugate gaps oppositely and cannot retain both. The exact finite dimer has both odd excitations; the exclusion concerns this projection regime, not every source or emergent sector. Finite gauge covariance also leaves operation parameters unspecified: two admitted rotations on one extended gauge source preserve all endpoint singlets but give complete target probabilities 16/25 and 144/169. A chosen simulator fixes its program; these gauge premises alone do not select its capacity-dependent action (`model/source_family_compatibility`, [1]).

The complete internal correspondence export matches all fifteen source weights, including color conjugation, before attaching Standard Model comparison names. Eighteen species or flavor channels retain distinct mass, family, defect and physical-identification fields. The three family names share one internal block until a source family map is supplied; neutrino flavor labels do not identify mass eigenstates. Physical source assignments and mass predictions are empty (`model/standard_model_source_map`, [1]).

For a protected defect to become an identified particle, one preparation must create its supported state, the same dynamics must preserve or control its charge while propagating it, and complete detectors must recover its energy, momentum and interactions. The memory charge above, the compact texture band and the gauge–Higgs response below have different carriers and actions. Their separate identities do not establish this common map.

### 5.16 A quartic interaction on the complete compact band

The spatial source is the cubic operator $h = T + Mn \cdot \sigma$, $\|T\| < 2$, $M \geq 8$. Its direction is the normalization of $(x, y, z^2 - a^2)$ inside the cube of radius $a + 1$ and a constant outside, with zero amplitude at the two cores $(0, 0, \pm a)$. The four-band bound is $r = b^2/(M - 4)$, where $b^2 = 7/4$ for $a = 1$ and $b^2 = 3/2$ otherwise. The spectral and particle–hole identities hold for this source by the compact-pair construction. Let $W$ contain the four localized modes and set $A = W^\dagger h W$. The isometry $E = W \otimes I_{45}$ embeds 180 annihilators into the complete finite field. The current and pairing intertwiners hold for every complete memory record, with $A$ retained in the quadratic core. Write $P = EE^\dagger$ and choose

$$V = \lambda\big(N_W - 90\big)^2, \qquad N_W = d\Gamma(P), \qquad \lambda \geq 0. \tag{12}$$

This is an explicit controller design. The particle–hole identity $C\overline{W} = WS$ ensures that the pairing has no localized/complementary cross term. All terms are even, so $H_r^{\text{full}} = H_{r,W} \otimes I + I \otimes H_{r,\perp}$ on every record. The interaction is parity even and commutes with all twelve native internal currents. In the localized occupation basis it contains the quartic term $2\lambda \sum_{i<j} n_i n_j$. The pairing Hamiltonian does not conserve $N_W$, and its commutator with $V$ need not vanish. Every record preserves the same localized algebra because $V$ acts entirely on that factor.

Hold the texture fixed throughout the memory averaging interval. The complete record limit has a bound independent of exterior volume. Writing the compact energies as $\{-e_2, -e_1, e_1, e_2\}$

with $e_i \le r$, the kinetic norm is $45(e_1+e_2) \le 90r$. If $g$ is the source feedback gain, the pairing bound and $0 \le N_W \le 180$ give

$$\|H_{r,W}\| \le M_W = 90r + 90(1+g) + 8100\lambda.$$

The radial two-column zero-mode construction is a separate special case with ninety annihilators, centre 45 and bound $45(1+g)+2025\lambda$. The actual memory transition has the reset decomposition $T = (1-p)R + pS$, and contracts zero-sum operator-valued memory blocks by at most $p$. At step size $t/n$, its reduced core state consequently satisfies

$$\|\rho_{W,n}(t) - e^{-it\bar{H}_W}\rho_W(0)e^{it\bar{H}_W}\|_1 \le \frac{4M_W t}{n(1-p)} + \left(\frac{8}{1-p}+4\right)\frac{M_W^2 t^2}{n}. \tag{13}$$

The estimate includes arbitrary initial classical–quantum memory correlations and a finite untouched reference. The first term vanishes for stationary product memory. The mean Hamiltonian contains the same memory-derived mass matrix and the interaction (12). Uniformity concerns this fixed localized algebra under its source isometries; it gives no uniform continuum estimate for all spatial observables.

Equation (13) compares the unread core. There is a separate estimate for the full native history register. For a fixed initial memory label, purify the core/reference state. Write $p_\omega$ for the field-independent history probability, $\psi_\omega = U_\omega\psi$, and $\bar{\psi} = e^{-it\bar{H}_W}\psi$. If $\delta$ is the unread trace-norm bound, then $\sum_\omega p_\omega(1-|\langle\bar{\psi}|\psi_\omega\rangle|^2) \le \delta/2$. The pure-state distance identity and Jensen's inequality bound the full history/core trace distance from $\sum_\omega p_\omega|\omega\rangle\langle\omega| \otimes |\bar{\psi}\rangle\langle\bar{\psi}|$ by $\sqrt{2\delta}$. Partial trace covers mixed inputs. For initial classical memory/core correlations, condition on the initial memory label and retain that tag in the ideal history law; convexity preserves the same bound. The complete final record/core rate is therefore $O(n^{-1/2})$. Opposite qubit phase records saturate this square-root relation. No uniform estimate for rare conditioned histories or intervening field measurements follows.

The interaction is defined through a spectral projector and need not be strictly onsite. For the radial special case, the resolvent estimate at texture coefficient 64 and $W_x^\dagger W_x = \rho_x I_2$ gives, for the projection $A_L$ onto $|x|_\infty < L$,

$$\|(I-A_L)E\|^2 \le \epsilon_L = \frac{3}{2\cdot 62^2}\,32^{-2(L-1)}.$$

Thus $\|P - A_L P A_L\| \le 2\sqrt{\epsilon_L}$, and finite rank gives $\|N_W - d\Gamma(A_L P A_L)\| \le 360\sqrt{\epsilon_L}$. The radial interaction $\lambda(N_W-45)^2$ is exponentially localized about the prescribed origin, without being a strictly onsite interaction.

On this radial subspace there is an exact obstruction for the onsite density alternative. On the two-particle sector let $D_g = \sum_x g_x \bigwedge^2 P_x$ and $E_2 = \bigwedge^2 E$. Direct compression gives

$$E_2^\dagger D_g(I - E_2E_2^\dagger)D_gE_2 = \left[\sum_x g_x^2\rho_x^2 - \left(\sum_x g_x\rho_x^2\right)^2\right] I. \tag{14}$$

Writing $c = \sum_x \rho_x^2 < 1$, Cauchy–Schwarz bounds the bracket below by $(1-c)\sum_x g_x^2\rho_x^2$. Every nonzero member of this onsite density class leaks from the source subspace on which it has support. This statement does not exclude a dressed low-energy subspace or other local interaction tensors.

**Connected dynamics and spectral shifts.** The compact mean pairing operator has 84 charged two-mode blocks and twelve unpaired neutrino modes. Select the two spatial pairs of one charged-lepton singular value $m$. Fix one occupation in each of the other 82 blocks and fill six neutral modes. Pair creation and annihilation kill these spectators, whose number is 88. Subtracting their fixed energy gives the exact reducing four-mode Hamiltonian

$$H_4 = e_1(n_1 - n_2) + e_2(n_3 - n_4) + m(a_1^\dagger a_2^\dagger + a_2 a_1 + a_3^\dagger a_4^\dagger + a_4 a_3) + \lambda(N_4 - 2)^2.$$

This is an invariant sector of the mean Hamiltonian, not its asserted vacuum or an invariant sector of every finite memory pulse. Pair parities split its sixteen levels into the four even/even values $0, 4\lambda, 2\lambda \pm 2\sqrt{\lambda^2 + m^2}$, four odd/odd values $\pm e_1 \pm e_2$, and eight mixed values $\lambda \pm e_i \pm m$. Consequently

$$E_0 = \min\{2\lambda - 2\sqrt{\lambda^2 + m^2}, -e_1 - e_2, \lambda - m - \max(e_1, e_2)\}.$$

This enumeration is a complete diagonalization by pair parity. For the unique odd/odd ground, a positive-energy orbital has two addition-plus-removal poles $e_1 + \lambda \pm m$, each of residue $1/2$. At $M = 64$, half-separation one and $\lambda = 1/128$, rational coefficient and exterior-volume enclosures place all three inherited charged-lepton slices in this ground sector:

| Quadratic coefficient $m$ | Lower pole | Upper pole |
|---:|---:|---:|
| 0.000002072030404 | 0.01171632730 | 0.01172047136 |
| 0.0004284302055 | 0.01128996912 | 0.01214682953 |
| 0.007205207119 | 0.004513192209 | 0.01892360645 |

The unit is the declared source pulse energy. These are slice poles, without a physical mass assignment or a determination of the full interacting vacuum. The compact exterior error $\delta_L$ bounds each displayed pole error by $\delta_L$ in this certified sector; no complementary mode is discarded.

The full 180-mode ground also has an explicit variational bracket. List the charged singular values, including colour multiplicities, as $m_j$, $j = 1, \ldots, 21$. Each occurs in two pairing blocks at each spatial energy $e_i$, with quadratic levels $\{\pm m_j, \pm e_i\}$. The twelve neutral modes contribute $-3(e_1 + e_2)$, giving

$$E_{\text{quad}} = -2\sum_{j=1}^{21}[\max(m_j, e_1) + \max(m_j, e_2)] - 3(e_1 + e_2).$$

Positivity of $V$ makes this a lower bound. A quadratic product ground has mean occupation 90 and number variance one per even pairing block, zero per odd block. If $b$ blocks are even, its trial energy is $E_{\text{quad}} + \lambda b$. A completely blocked trial state has exact occupation 90, zero interaction energy and kinetic energy $-45(e_1 + e_2)$. Thus, retaining every localized mode,

$$E_{\text{quad}} \le E_{\text{full}} \le \min\{E_{\text{quad}} + \lambda b, -45(e_1 + e_2)\}.$$

The specified coefficient witness has $b = 28$. These bounds constrain the full vacuum energy without identifying its excitation spectrum.

The radial control sets $e_1 = e_2 = 0$. Its ground energy is $2\lambda - 2\sqrt{\lambda^2 + m^2}$, and its single-mode spectral poles and weights are

$$\omega_\pm = 2\sqrt{\lambda^2 + m^2} - \lambda \pm m, \qquad w_\pm = \frac{1 \mp m/\sqrt{\lambda^2 + m^2}}{2}.$$

In this radial control the cross-pair density cumulant is $-\lambda/(4\sqrt{\lambda^2 + m^2})$, while its normal and anomalous cross-pair two-point contractions vanish. Hence positive $\lambda$ gives a non-Gaussian connected interaction. These spectral shifts preclude identifying the quadratic mass coefficients with the interacting excitation energies without further calculation.

**A common finite energy and modular response.** For the same full localized Hamiltonian choose $\rho_\beta = Z^{-1}e^{-\beta\bar{H}_W}$, $\beta > 0$. This state is faithful. A global spectral reset with Kraus operators $\sqrt{p_i}|i\rangle\langle j|$ prepares it and preserves total probability. That reset is a declared service, not an implementation by the original site repair. For $H(s) = \bar{H}_W + sX$, Duhamel differentiation gives

$$\rho'_\beta(0) = -\beta \int_0^1 \rho_\beta^u (X - \langle X\rangle)\rho_\beta^{1-u}\, du.$$

The positive logarithmic-mean kernel yields $F' = \langle X\rangle$, $F'' = -\beta C_{\rho_\beta}(X, X) \le 0$, and $S' = \beta(E' - \langle X\rangle)$, where $C_\rho$ is the Bogoliubov–Kubo–Mori covariance. The noncommuting choice $X = (N_4-2)^2$ is checked against an independent matrix-exponential derivative. These are finite identities for one nonquadratic Hamiltonian. They supply no geometric modular flow, local relativistic stress, area normalization or geometric entropy stationarity, and therefore imply no Einstein dynamics. Exact characteristic polynomials, complete finite spectra, record controls and the onsite leakage calculation accompany the proof.

**Intervening observations.** The bounds above extend to adaptive experiments. On the compact 180-mode core at $\lambda = 1/128$, every record's localized Fock generator obeys $\|H_{r,W}\| \le M_W \le 7869/32$. Between $m$ evolution intervals, apply arbitrary complete quantum instruments to the field and any auxiliary observer register, chosen from every retained native and detector record, with all failure outcomes kept. Replacing actual interval channels by ideal mean-dynamics channels one at a time bounds the complete-history distance by $\sum_j \min\{2, \sqrt{2\delta_j}\}$, where $\delta_j$ is the unread bound of interval $j$. The native memory continues under its actual transition law, since conditioning does not justify resetting it to a stationary product. Exact resource schedules and a complete finite execution check the correspondence. The bounds concern a declared temporal refinement, and adapting at every microscopic step needs its own budget.

### 5.17 Work on the moving compact interacting band

A deformation must act on the same Hamiltonian used to identify the matter states. On the compact opposite-degree texture, vary the actual hopping by $h_s = sT + MV$, with $1/2 \le s \le 3/2$, $M \ge 8$ and $\|T\| < 2$. The texture, memory coefficients and quartic coupling are fixed. The four-dimensional spectral band has energies in $[-3, 3]$ and separation at least $M - 6$ from its complete complement. Let $P_s$ be its projector, $\mathsf{C}$ the source particle-hole matrix, and $K_\omega$ the internal pairing kernel on a complete memory record $\omega$. The same source law is

$$N_s = \mathrm{d}\Gamma(P_s \otimes I_{45}), \qquad H_{\omega,s} = \mathrm{d}\Gamma(h_s \otimes I_{45}) + \mathrm{pair}(\mathsf{C} \otimes \overline{K_\omega}) + \lambda(N_s - 90)^2.$$

Its exact force conjugate to the hopping control is

$$\partial_s H_{\omega,s} = \mathrm{d}\Gamma(T \otimes I_{45}) + \lambda\{N_s - 90, \mathrm{d}\Gamma(P'_s \otimes I_{45})\}. \tag{15}$$

The second term follows by differentiating the noncommuting square. Freezing the localization projector omits this contribution to source work.

**Proposition 5.2** (Localized force with the moving band retained)**.** *The band admits a unitary comparison frame $W_s$ in which its pairing form is constant and its intrinsic force is*

$$J_s = \mathrm{d}\Gamma((W_s^* T W_s) \otimes I_{45}), \qquad \|J_s\| \le 360, \qquad \|P'_s\| \le \frac{6}{M-6}.$$

*For two even localized states with the same even complement reference, their contrast of* (15) *equals their contrast of $J_s$. The bounds are independent of the containing finite cube.*

*Proof.* For band vectors $p$ and complementary eigenvectors $q$, $\langle q, P_s' p\rangle = \langle q, Tp\rangle/(E_p - E_q)$ and the diagonal blocks of $P_s'$ vanish. Hence $\|P_s'\|^2 \le \|P_s'\|_{\mathrm{HS}}^2 \le 8\|T\|^2/(M-6)^2 < 36/(M-6)^2$. Set $L_s = [P_s', P_s]$ and solve $U_s' = L_s U_s$. Then $P_s = U_s P_1 U_s^*$ and $W_s = U_s W_1$. Particle-hole covariance gives $\mathsf{C}\overline{U_s} = U_s\mathsf{C}$, so the pairing form is fixed. Differentiating $A_s = W_s^* h_s W_s$ gives $A_s' = W_s^*(T + [h_s, L_s])W_s = W_s^* T W_s$: $h_s$ is block diagonal and $L_s$ off diagonal. Number and pairing therefore have no derivative in this transported frame. The 180-mode second-quantization norm is at most $180\|T\| < 360$. In an even product state the cross-band terms have zero expectation; subtracting the same complement reference cancels its remaining energy. □

At fixed $s$, a complete-memory trace-norm error $\delta$ consequently gives a force-readout error at most $360\delta$. The comparison transporter is an explicit mathematical identification; Hamiltonian evolution with changing $s$ does not implement it automatically. For the supplied Gibbs reference of the same mean Hamiltonian at fixed $\beta > 0$, Duhamel differentiation gives

$$F'_{\mathrm{thermal}} = \mathrm{Tr}(\rho_s J_s), \qquad S' = \beta\big(E' - \mathrm{Tr}(\rho_s J_s)\big).$$

The complete compact source and a reducing sixteen-state interacting slice check these derivatives, retaining the spatial splitting and work performed by the hopping control. The hopping amplitude is a declared control. Its work derivative supplies no identification with a metric variation or conserved Lorentz stress, and this finite Gibbs identity supplies no regional geometric modular flow, area normalization or fixed-volume generalized-entropy stationarity.

### 5.18 A finite gauge–Higgs response with complete detector records

A separate two-vertex control joins leptonic fermion hopping, two oppositely oriented native 60-state gauge links and the charged-lepton Higgs tensor. There are three families and two spin components. The scalar at vertex zero has vacuum or one conjugate-doublet quantum; colored fermions have zero occupation and the up/down vertices are disabled. Colored link sectors are retained. The conserved quantity is $Q_* = N_F - 2N_H$, and Gauss constraints imply $N_L = N_H$, $N_e = Q_* + N_H$ and bulk hypercharge $6Y = 6Q_*$. A reference of charge $-6Q_*$ makes the complete state neutral. The scalar cap defines this finite model, rather than an invariant truncation of an unrestricted bosonic field.

For $F$ families the complete physical dimensions are

$$d_0 = 6 + 48F^2, \qquad d_1 = 16F + 64F^3 - 20F^2.$$

At $F = 3$ these are 438 and 1596. Exact Gauss kernels, independent representation counts and the full operator images verify completeness and zero complement leakage. The charged scalar-vacuum subspace has dimension 48, with its entire 1548-dimensional occupied complement retained. Writing $A = PHP$, $C = QHP$ and $D = QHQ$ gives

$$P(z-H)^{-1}P = [z - A - \Sigma(z)]^{-1}, \qquad \Sigma(z) = C^\dagger (z-D)^{-1} C.$$

For $\mathrm{Im}\, z > 0$, $-\mathrm{Im}\,\Sigma(z) \succeq 0$. In the declared unit-strength control, $\|\Sigma(1+0.5i)\| \simeq 1.793879153064$ and the hopping/Higgs commutator is nonzero. This verifies interaction in the finite system.

Let $H_0\Omega = E_0\Omega$ be its prepared neutral ground and $b_j = c_j^\dagger d_{\mathrm{ref}}^\dagger$ the six family/spin creation maps. The odd reference makes $b_j$ even. Each is a partial isometry, and the full signed response is

$$C_{jk}(t) = \langle \Omega | b_j^\dagger e^{-i(H_1 - E_0)t} b_k | \Omega\rangle = \sum_n e^{-i(E_{1n} - E_0)t} \langle b_j\Omega | n\rangle\langle n | b_k \Omega\rangle.$$

Every spectral row and zero residue is retained. For the specified family matrices, unit dynamical strengths and scalar energy, and zero fermion/reference energies, the first three distinct numerical addition frequencies are $2.58004293348 \times 10^{-8}$, 0.001006653510723 and 0.452014700447579. Exact-input 50/80-digit residual refinement supports the tiny positive first difference; it supplies no interval-certified enclosure or ordering.

These frequencies enter a complete measurement, not a postselected creation curve. On the direct sum of the two sectors set

$$G_j = \begin{pmatrix} 0 & b_j^\dagger \\ b_j & 0 \end{pmatrix}, \qquad W_{s,j} = I - G_j^2 - isG_j, \qquad s = \pm 1.$$

Since $G_j^3 = G_j$, $W_{s,j}$ is unitary. With $U(t) = e^{-it(H_0 \oplus H_1)}$, a detector ancilla compares the forward paths $W_{+,j}U(t)$ and $U(t)W_{s,k}$. Their overlap is $F_{s,jk} = A_{jk} + sC_{jk}$, where $A_{jk} = \langle a_j | e^{-i(H_0 - E_0)t} | a_k \rangle$ and $a_j = (I - b_j^\dagger b_j)\Omega$ retains blocked creation. The two detector bases give

$$p_\pm^X = \tfrac{1}{2}(1 \pm \operatorname{Re} F_{s,jk}), \qquad p_\pm^Y = \tfrac{1}{2}(1 \pm \operatorname{Im} F_{s,jk}), \qquad C_{jk} = \tfrac{1}{2}(F_{+,jk} - F_{-,jk}).$$

The maps $(W_{+,j}U \pm UW_{s,k})/2$ and $(W_{+,j}U \mp iUW_{s,k})/2$ retain both outcomes and their back-action on arbitrary inputs, including purifications. The controlled ancilla, quarter-turn probes and ground preparation are supplied services.

For independent prepared shots at times $t_i$, compare $H_1$ with $H_1 - \delta\Pi$, where $\Pi$ is a spectral projector. The two pre-readout pure states have overlap $z = 1 + (e^{i\delta t} - 1)w/2$, $w = \|\Pi b_k \Omega\|^2 \le 1$, hence $1 - |z|^2 \le \delta^2 t^2/4$. Multiplication of fidelities and contraction under the complete measurement give the record-only bound

$$\mathrm{TV} \le \min\left\{1, \frac{|\delta|}{2}\sqrt{\sum_i t_i^2}\right\}.$$

The supplied schedule has $24 \times 4096$ shots at each $t = i/10$, $0 \le i \le 128$: 12,681,216 retained bits. For $|\delta| \le 3 \times 10^{-8}$ the bound is strictly below $1/2500$. An observer who reads the simulator specification can have further information excluding that comparison Hamiltonian. Applying this bound to an exactly vanishing smallest line also requires an enclosure absent from the numerical precision ladder.

Finally, $\epsilon_F N_F + \epsilon_H N_H = \epsilon_F Q_* + (2\epsilon_F + \epsilon_H)N_H$. Changing $(\epsilon_F, \epsilon_H)$ by $(\delta, -2\delta)$ shifts addition frequencies if the reference energy is fixed. A common gauge-potential shift changes the compensating reference by $-\delta$ and cancels exactly in the complete Hamiltonian. This is a reference-calibration distinction, not freedom to fit intrinsic masses. The graph, scalar cap, couplings, preparation and physical clock are not selected by these finite identities. Full certificates and record instruments are in the repository package `model/interacting_extension` [1].

**Native state and carrier bounds.** The 66-dimensional occupation factor $\rho_*$ of Section 5.15 has eigenvalues 625/7505, 200/7505 and 64/7505, with multiplicities $1, 20, 45$. Its ideal modular commutator, divided by $\log(25/8)$, obeys $D(D^2 - I)(D^2 - 4I) = 0$. Any injective target-algebra map into this source that intertwines the two derivations must preserve that polynomial identity. This is a condition on an exact common clock, without a numerical exclusion of the Higgs response lines. The full $\rho_\alpha$ also carries its reference-dependent memory gap.

An isometric image of $\rho_*$ retains its 66 nonzero eigenvalues, so it has trace distance at least $1 - 625/7505 = 1376/1501$ from every pure target. A pure field subsystem is possible if other retained quantum memory carries the mixedness; its preparation and records must then be supplied.

Separately, if an entire quantum channel crosses a single 66-dimensional carrier, arbitrary classical records cannot raise its entanglement fidelity above $66/D$ on a $D$-dimensional input. With no additional quantum path or entanglement assistance, its half-diamond distance from the identity is therefore at least $62/73$ for $D = 438$, or $255/266$ for $D = 1596$. The bound concerns all inputs and references, rather than one prepared ground-state experiment. The named observer's coherent work register is an additional quantum resource. These exact tests rule out direct identifications while leaving explicit active preparations and larger encodings to their own comparison theorems (`model/source_state_bridge`, [1]).

### 5.19 Compensated probe uncertainty and collision records

The six creation maps above also supply an exact uncertainty relation on the complete finite carrier. For any one map $b$, set

$$P_0 = b^\dagger b, \quad P_1 = bb^\dagger, \quad P = \mathrm{diag}(P_0, P_1), \quad D = \mathrm{diag}(P_0, -P_1),$$

$$X = \begin{pmatrix} 0 & b^\dagger \\ b & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -ib^\dagger \\ ib & 0 \end{pmatrix}.$$

Each normalized $b$ is a signed partial permutation of rank 384. Consequently $X^2 = Y^2 = D^2 = P$, $[X, Y] = 2iD$ and $\{X, Y\} = 0$. For either $A = X$ or $Y$, the spectral projectors $(P - A)/2$, $I - P$ and $(P + A)/2$ have ranks 384, 1266 and 384. The zero outcome retains 54 neutral blocked states and 1212 charged states outside the range. These identities are checked in exact rational complex arithmetic for all six source maps.

For any density supported on the neutral sector, including a reduced density entangled with an arbitrary reference, let $w = \mathrm{Tr}(\rho P_0) = 1 - \langle n_j \rangle$. Both means vanish, both variances equal $w$, and

$$(p_{-1}, p_0, p_{+1}) = (w/2, 1 - w, w/2), \qquad \Delta X \, \Delta Y = w = \tfrac{1}{2} |\langle [X, Y] \rangle|.$$

Thus the supplied finite density saturates Robertson's inequality [18]. The tracial neutral state gives $(32/73, 9/73, 32/73)$. The complete maps $\rho \mapsto E_a \rho E_a$ preserve the untouched reference marginal when outcomes are summed. These are exact projective instruments, distinct from the two-outcome Ramsey readout; a native pulse compiler for them is not asserted. The density and complex measurement semantics are those of the specified finite Higgs source.

These even charge-compensated quadratures do not identify physical position and momentum. Finite matrices satisfy $\mathrm{Tr}[Q, P] = 0$, hence $\|[Q, P] - i\hbar I\| \geq |\hbar|$ for nonzero $\hbar$. The source's separate fluctuation and ordered-spin constructions obtain canonical field limits on their stated states and domains; the number of independently prepared fluctuation copies is not the physical memory capacity. The ordered field's Gram form constructs a position conjugate to its existing spatial translation generator, on an explicit field-vector domain. A complete extension includes a no-position outcome. Declared coherent source services implement complete finite detectors, and finite Gram compressions give ordered convergence on fixed states and references. This gives no effective source-size modulus or moment-tail bound. A larger response space, generated by all time-evolved field vectors, admits a covariant position measure and a time-invariant dark complement. It need not be a particle sector. Protected total-load coordinates are scalar and cannot supply that position. Finite uncertainty alone selects no Planck scale or particle interpretation [6]. The proof, full instruments and source bindings are in the `model/particle_identification` package [1].

**Capacity and complete local experiments.** The native interpreter has an exact record-capacity result. Under its `export_oldest` policy, capacities one through eight give the same complete experimental channel after reversible resegmentation of archive and queue, with carrier, commands, parameters and evaluator limits fixed. Each adaptive selector reads only the last retained record; export preserves it. Induction over raw branches therefore gives zero channel distance, including arbitrary references and resource-stop outcomes. The archive is part of the output; readable capacity specifications and resource costs differ. With the `halt` policy, two holds at capacities one and two instead have disjoint timeout/completion statuses. This theorem concerns record slots, not physical global capacity or a capacity-dependent particle action; its proof and exact controls are in `model/particle_identification/NATIVE_CAPACITY.md` [1].

For the specified full-Fock Weyl/contact action at fixed spacing, nested local buffers give a stronger comparison than agreement of a few frequencies. Let $\epsilon_0$ bound the initial trace distance of the joint largest-buffer, observer and reference state. At stage $i$, let $\delta_i$ be the complete operator locality error, $\eta_i$ the unitary splitting error, and $\zeta_i$ the half-diamond error of the complete instrument, uniform over incoming states, references and previous records. For target $(t_i, c, U)$ and simulator $(s_i, c', U')$, let $E_i, V_i$ count buffer edges and sites. Then

$$a_i = E_i|t_i c - s_i c'| + V_i|t_i U - s_i U'|, \qquad D_{\text{final}} \le \min\left\{1, \epsilon_0 + \sum_i (\delta_i/2 + \eta_i + a_i + \zeta_i)\right\}.$$

Here $D_{\text{final}}$ is trace distance on the entire retained local quantum state and classical record. Each edge has two hopping terms of norm $c/2$; each site has a contact projector. Duhamel's formula bounds the local action change by $a_i$, and complete-channel contraction telescopes all stages, including adaptive instruments and failure records. Durations are fixed by stage. The bound has no ambient-volume factor, but comparison across physical capacities requires a common local identification and controlled couplings, clock, joint input and instruments. No such capacity-to-action map follows from volume independence. Nor does convergence at fixed time imply control of arbitrarily small spectral gaps.

**Spatial scattering and physical identification.** The contact source has complete two-body lattice wave operators and a unitary scattering matrix on its absolutely continuous subspace. Critical tuning of the moving-contact family gives a nontrivial two-body continuum limit in three dimensions, with all eight edge valleys retained. This limit is nonrelativistic. Its separated bound pairs prevent identifying the empty additive Fock state as a ground vacuum. The accompanying packet construction controls incoming and outgoing time tails, packet truncation, preparation, finite volume, splitting and native readout, and separates a retained interaction event from its matched free law by more than 3/10. That fine-spacing finite schedule is constructive; the executed five-qubit control is an interaction echo, not an executed asymptotic collision. The respective packages are `model/contact_scattering`, `model/moving_contact_scattering` and `model/scattering_packet_readout` [1].

A collider identification requires one source-selected state, spatial action, clock and detector family to join these ingredients. Separated incoming packets, their interacting propagation and complete outgoing bins must share quantitative errors, including soft and failed events. The effective theory must control its relevant relativistic symmetries, regulator sectors and resolution window. An infinite ultraviolet limit is not required merely to compare a finite-resolution experiment, but a free dispersion relation cannot replace the interacting record law. Stable neutral shells, charged inclusive thresholds, resonance line shapes and scheme-dependent confined parameters then require their respective physical definitions, as described below.

# 6 Comparison with the measured spectrum

## 6.1 Observables and conventions

At finite volume the spectral measure of a prepared operator $O$ is

$$d\mu_O(\omega) = \sum_n |\langle n|O|\Omega\rangle|^2 \delta(\omega - (E_n - E_\Omega))\, d\omega.$$

A visible eigenvalue requires nonzero residue in the actual detector channel, and the physical dictionary must transport $O$, the state, the Hamiltonian, current charges and the clock together. Localized wavefunctions are not momentum eigenstates, and a Fourier transform of a pinned mode is no moving-defect dispersion experiment.

In a justified relativistic regime a stable massive excitation has a common rest-energy and propagation relation. An unstable particle requires a resonance convention and width [15]. Charged sectors coupled to a massless photon require an infrared treatment rather than an assumed isolated sharp pole [17, 20], and color confinement prevents reading a free color-mode frequency as an observable quark or gluon mass. Running charged masses and mixing data must share scale and scheme [13, 14]. The finite source spectra of Section 3 have no certified running-to-pole, confinement, or laboratory-clock map, and they enter no row of the table below.

## 6.2 The twelve comparison values

The comparison branch is specified by a boundary packet: pixel coordinate $P = 1.630971465307864$, unified coupling $\alpha_U = 0.04112426184$, external unit $E_* = 1.22089 \times 10^{19}\,\mathrm{GeV}$, vacuum expectation value $v = 246.60010\,\mathrm{GeV}$ and scale $Q_0 = 91.58958701\,\mathrm{GeV}$. Its $W$ and $Z$ couplings use one-loop supersymmetric running, its Higgs and top values use imposed ultraviolet criticality, and its quark Yukawa couplings are the six-power texture

$$Y_u = \mathrm{diag}(6^{-6}, 6^{-3}, 1), \qquad Y_d = \mathrm{diag}(6^{-6}, 6^{-4}, 6^{-2}).$$

No proof here selects these values from finite repair. Multiloop outputs come from the pinned Standard Model in Dimensional Regularization code [12], with two corrections to its electroweak decoupling routine, the threshold-function operator and the placement of the strong-coupling assignment. A replay reproduces the five light quarks and the $W$, $Z$, Higgs and top to maximum relative discrepancy $5.85 \times 10^{-11}$. Table 3 gives the result against the 2026 Particle Data Group tables [16].

For positive outputs $x_i$ and intervals $[l_i, u_i]$, one common factor is feasible exactly when $\max_i l_i/x_i \le \min_i u_i/x_i$. The intersection is empty for all twelve values, for the charged triple alone and for $W, Z$ alone. This is an exact statement about the printed rectangle.

## 6.3 Charged leptons

The lepton circulant is chosen near the measured pole-mass Koide coordinates. Its response constants $50, 31, 310, 512, 77, 21, 27, 135$, its determinant exponent and the pole reading are target-informed: with the measured electron or muon mass the law returns $m_\tau = 1776.93\,\mathrm{MeV}$, within $2 \times 10^{-3}$ of the world-average uncertainty, and this agreement carries no evidential weight. Absolute values use the external unit $E_*$. Summing the three cosines and their squares proves $Q = \sum_j m_j/(\sum_j \sqrt{m_j})^2 = (1 + e^{-2\chi})/3$, and the packet has $Q = 0.66666446$, $\rho = \sqrt{2}e^{-\chi} = 1.41420889$ and $\delta = 2/9 + \zeta = 0.22222476$, against the same coordinates of the measured pole triple to nine digits.

Table 3: The twelve nonzero comparison values, in GeV. All twelve lie outside their printed intervals. No covariance-aware joint significance or total theory-error bound is asserted.

| | Output | Reference | Residual | Convention |
|---|---|---|---|---|
| $e$ | 0.000510962911 | 0.00051099895069 | −70.53 ppm | Pole premise |
| $\mu$ | 0.105650931342 | 0.1056583755 | −70.45 ppm | Pole premise |
| $\tau$ | 1.77680479525 | 1.77693 | −70.46 ppm | Pole premise |
| $u$ | 0.00668144352 | 0.00216 | +209.33% | $\overline{\text{MS}}$, 2 GeV |
| $d$ | 0.00665065487 | 0.0047 | +41.50% | $\overline{\text{MS}}$, 2 GeV |
| $s$ | 0.239423082 | 0.0929 | +157.72% | $\overline{\text{MS}}$, 2 GeV |
| $c$ | 1.56844907 | 1.2729 | +23.22% | $\overline{\text{MS}}$ self-scale |
| $b$ | 6.59077908 | 4.186 | +57.45% | $\overline{\text{MS}}$ self-scale |
| $t$ | 170.611817 | 172.1 | −0.865% | Pole, cross-section comparator |
| $W$ | 79.8563905 | 80.3625 | −0.630% | Breit–Wigner |
| $Z$ | 90.5845168 | 91.1879 | −0.662% | Breit–Wigner |
| $H$ | 124.842549 | 125.13 | −0.230% | Pole |

Reading the outputs as pole masses is an adopted modeling premise. Under it the discrepancy is almost entirely a common scale of about −70.5 parts per million. The muon/electron ratio residual is +0.0725 parts per million and the tau/electron ratio residual is +0.0662 parts per million. A common normalization cannot remove the first, and the electron and muon intervals admit no shared factor. The exact Koide point $\rho = \sqrt{2}$, $\delta = 2/9$ gives $m_\mu/m_e$ off by +9.83 parts per million and $m_\tau/m_e$ off by +30.9 parts per million; its tau ratio lies in the printed rectangle and its muon ratio does not. Assigning the same outputs running masses at $Q_0$ and applying the retained one-loop matching relation instead shifts the pole comparisons by $(4.854, 2.870, 1.820)$ percent. Proximity to pole coordinates selects neither scheme.

### 6.4 Quarks

The declared texture has the transported light ratios $m_s/m_d \simeq 35.99993$ and $m_u/m_d \simeq 1.00463$, in the modified minimal subtraction scheme with quantum chromodynamics and electrodynamics, four active quarks and three charged leptons at 2 GeV. The rectangle of the two quoted strange and down intervals is approximately $[19.33, 20.22]$, and the directly quoted $m_u/m_d$ interval is $[0.442, 0.482]$. A common multiplier cancels in each ratio, so no energy unit can repair the light-quark shape, and this texture is rejected. Precise identification with the reference quantum-chromodynamic masses also needs an electromagnetic scheme conversion.

For a coefficient $m = A\,6^{-n}$ and ratio $r =$ output/reference, holding the transport factor fixed gives $n_{\text{required}} = n + \log_6 r$. The $u, d, s, c, b$ coordinates are approximately 6.63023, 6.19375, 4.52837, 3.11653 and 2.25334. Self-scale masses and threshold dependence require a coupled rerun for an actual changed boundary, so these are diagnostic coordinates and no selected replacement texture. Matched five-flavor running gives $\alpha_s(91.1879\,\text{GeV}) = 0.11944066$ for the stipulated branch; a reference-ratio estimate of 0.11945742 is a different operation.

### 6.5 Weak bosons, Higgs and top

The $W, Z, H, t$ residuals depend on the renormalization convention and attachment scale of the supplied vacuum expectation value. The source value agrees with the reference model's $v(172.4\,\text{GeV}) = 246.60030\,\text{GeV}$ within one part per million, while the branch attaches it at $Q_0$, where the reference

value is 248.38564 GeV. Table 4 substitutes reference inputs at $Q_0$ to diagnose this sensitivity. A substitution derives no attachment, and the numerical equality supplies no scale-selection theorem. The controls omit the stored finite-light shifts of 1.7 and 13 keV.

Table 4: Percent residuals under reference substitutions at $Q_0$.

| Inputs replaced by reference values | $W$ | $Z$ | $H$ | $t$ |
|---|---|---|---|---|
| None | −0.630 | −0.662 | −0.230 | −0.865 |
| Gauge couplings | −0.717 | −0.698 | −0.241 | −0.899 |
| $v$ | +0.076 | +0.047 | +0.436 | −0.200 |
| $v$ and gauge couplings | −0.012 | +0.011 | +0.425 | −0.233 |
| $v$, $y_t$, $\lambda$ | +0.063 | +0.029 | +0.110 | +0.245 |
| All running inputs | −0.025 | −0.008 | +0.099 | +0.210 |

The weak vectors use the stated Breit–Wigner convention, and the top comparison uses the $172.1 \pm 0.6$ GeV cross-section pole extraction, distinct from a direct event-generator mass. Loop-order truncation, reference-input uncertainties and physical model error are distinct from numerical replay precision, and criticality itself is sensitive to loop order: at two loops the replayed boundary has $\lambda = 2.3 \times 10^{-8}$, and at three loops it has $\lambda = 9.7 \times 10^{-4}$.

### 6.6 Massless and neutral sectors

The photon and gluon zeros are conditional action coefficients. The minimal neutrino action gives three zero masses and zero oscillation gaps, contrary to the nonzero measured gaps, and individual absolute neutrino masses are unmeasured. The specified unshifted amplitude-to-mass attachment is excluded by Proposition 7.3. No particle abundance or early-Universe population is computed by this branch.

These are conditional comparisons, including target-informed choices. None of these constructions is a derivation of the measured spectrum. The finite gauge and matter results select no continuous mass spectrum, mixing matrices or population history. A joint physical test requires one source law, its state and complete record map, compatible mass conventions and declared experimental and theoretical uncertainties. Exact equality to measured central values is not an experimental theorem; equality of observable laws is the stronger mathematical target.

## 7 What records and symmetries exclude

Each statement below is exact on its stated domain and removes a tempting route to a spectrum. None excludes OPH as a whole; together they locate the missing selection.

### 7.1 Coherent phases and family averaging

The native event rates $p = 1/12$, $q = 1/6$ satisfy $2\sqrt{p/q} = \sqrt{2}$ and $p + q - 2pq = 2/9$, the base coordinates of the charged circulant.

**Proposition 7.1** (Classical records select no coherent phase)**.** *For a finite instrument with Kraus operators $K_{ra}$ and arbitrary real $\theta_{ra}$, replacing $K_{ra}$ by $e^{i\theta_{ra}}K_{ra}$ leaves every outcome map, every repeated composition and every record-dependent feedback experiment unchanged, on the full matrix space and with arbitrary entangled references. A well-defined function of the instrument maps therefore cannot depend on these phases.*

A coherent sum $\sum_{ra} c_{ra}K_{ra}$ thus does not follow from the instrument alone, and setting a probability equal to an angle in radians is an additional rule. An explicit same-rate counterfamily, separate from the native 128-record instrument, has identical classical instruments and different coherent spectra. A source-selected Stinespring isometry with an accessible coherent reference can carry the missing information, and such a reference is outside this exclusion.

**Proposition 7.2** (Real cyclic averaging forces a repeated eigenvalue)**.** *For a real regular three-cycle $R$ and $P = (I + R + R^2)/3$,*

$$\tfrac{1}{3}\sum_{j=0}^{2} R^j X (R^T)^j = aP + b(I - P), \qquad X = X^T \in M_3(\mathbb{R}).$$

The two-dimensional rotation plane forces the repeat, so cyclic averaging of a real symmetric kernel cannot select three distinct family values. An equivariant construction from real inputs has real outputs, and a nonreal chiral response needs an input or selection rule supplying the oriented complex structure.

### 7.2 The passive family metric

For the native core-biased reference and nonnegative seam weights, stationarity gives $\sigma^T(I-T)\sigma = (157/90000)W_{\mathrm{cross}}$, which forces all eight core-to-outer weights to vanish. Every retained branch conserves a nonzero core-charge coordinate. The harmonic extension $U$ of the native passive Schur calculation obeys $B^TU = GI$ and

$$Z = U^TU/G = I + (U - B)^T(U - B)/G \geq I.$$

The Schur mass matrix has the conserved family direction as a null vector, and the maximal 22-seam mixture attains rank two. Positive normalization is an invertible congruence and preserves rank, so it cannot generate the third charged family coefficient. A different reference or active recovery changes the premises of this obstruction.

### 7.3 Neutral hierarchy

**Proposition 7.3** (Neutral squared-gap bound)**.** *Put $z = e^{-ih}$, $U = I+(z-1)P_2$, $V = I+(z-1)P_3$ on the bright plane of the two native rank-one family projectors, which have $\mathrm{tr}(P_2P_3) = 1/5$, and let $F = \left(\begin{smallmatrix}(1-p)U & U\\ pV & 0\end{smallmatrix}\right)$ be the recurrent transfer. For $0 \leq p \leq 1/12$, $0 < h \leq 1/2$, its two roots of modulus above $1/2$ have phase coefficients $\omega, 1-\omega$ with*

$$0 \leq \omega \leq a = \tfrac{1}{2}\left(1 - \tfrac{95}{96}\sqrt{\tfrac{653}{848}}\right).$$

*Under the common-scale attachment $(m_1, m_2, m_3) = M(0, \omega, 1-\omega)$, the adjacent squared-gap ratio is at most $a^2/(1-2a) < 0.004988$.*

The determinant of $F$ reduces to a four-dimensional calculation, with the other ten memory inputs retained as a transient block of induced norm at most $p$. Substituting $\lambda = e^{-ih/2}x$ and $y = x - p/x$ reduces the quartic to $y = (1-p)\cos(h/2) \pm i\sqrt{C_0}\sin(h/2)$, with $C_0 = (1-p)^2 + 4p/5$, and elementary bounds on the slow root give the estimate. The measured normal-ordering ratio is 0.03108, with printed rectangle $[0.03011, 0.03208]$, which excludes this attachment. Arbitrary neutral actions, offsets and other mass readouts are outside the proposition, and it fixes no absolute neutrino mass.

### 7.4 Residues, references and mass readouts

The complete instrument determines which spectral values an observer can read. Let

$$T = \begin{pmatrix} 1-p & 1 \\ p & 0 \end{pmatrix}, \qquad \pi = \frac{(1,p)}{1+p}, \qquad U_j = \exp(-ihX \otimes P_j),$$

with $0 \le p \le 1/12$. On the bright plane $B$, choose its antisymmetric area form $J$ and put $A = |+\rangle\langle -| \otimes J$, $O = A + A^*$, $P_B = I_2 \otimes B$. Then $U_j A U_j^* = e^{-ih} A$ and $O^2 = P_B$. The three projectors $(P_B + O)/2$, $(P_B - O)/2$ and $I - P_B$ give a complete readout with values $+1, -1, 0$. Starting from $\pi$ and the stationary bright, dark or tracial response state $P_B/4$, $(I - P_B)/2$ or $I_6/6$ gives, respectively,

$$C_B(n) = \cos(nh), \qquad C_D(n) = 0, \qquad C_{\mathrm{tr}}(n) = \tfrac{2}{3}\cos(nh).$$

The same transfer frequency therefore has a nonzero or a zero measured residue depending on the supplied state and detector, with the first measurement operation included.

Density records also cannot fix absolute field frequencies. Adjoin a reference line and extend each branch by $V_j(\delta) = 1 \oplus e^{-i\delta h} U_j$. All complete record probabilities and conditional density matrices confined to the original response sector are independent of $\delta$, while coherences with the added reference acquire that phase, $F_\delta = e^{-i\delta h} F_0$.

A coherent family potential need not be a scalar mass. For the supplied Dirac symbol $\alpha_i = Z \otimes \sigma_i \otimes I$, a potential $I_4 \otimes A$ has zero projection onto the spin-invariant terms anticommuting with all $\alpha_i$. For Hermitian $A, M$ and $H_h(k) = \alpha \cdot k + X \otimes I_2 \otimes M + hI_4 \otimes A$,

$$\mathrm{Tr}[(H_h(k)^2 - |k|^2 I)^2] - \mathrm{Tr}[H_h(0)^4] = 16h^2|k|^2 \,\mathrm{Tr}(A^2),$$

so a nonzero potential in this truncated Hamiltonian is no collection of constant relativistic masses. Finally, cyclic trace gives $\mathrm{Tr}([K,P]H) = \mathrm{Tr}(K[P,H])$. If $[P,H] = 0$, $[K,P] = iH$ and $H$ is Hermitian, then $\mathrm{Tr}(H^2) = 0$ and $H = 0$. Exact global finite translation and boost identities therefore require a different domain from a nonzero relativistic Hamiltonian; controlled sectors and infinite limits carry their own error bounds.

### 7.5 Comparing capacities

Finite tests can compare whole record laws without enumerating histories.

**Proposition 7.4** (Sufficient criterion for equal record laws)**.** *Let $A_y, B_y$ be nonnegative outcome matrices of two finite classical instruments with column-stochastic sums, and $C$ a nonnegative column-stochastic map. If $CA_y = B_y C$ for every record $y$ and $Cp = q$, then all finite joint record laws coincide. With $\epsilon_0 = \|Cp - q\|_1$ and $\delta = \max_j \sum_{y,i} |(CA_y - B_y C)_{ij}|$, the record total-variation distance after $n$ steps is at most* $\min(1, (\epsilon_0 + n\delta)/2)$.

Induction gives the exact case, and a telescoping sum of $\ell_1$ contractions gives the approximate one. Equality of unread channels is insufficient: the one-state instruments $A_0 = 1, A_1 = 0$ and $B_0 = 0, B_1 = 1$ have the same unread channel and disjoint records. Determinism supplies neither the intertwiner nor a compatible initialization.

# 8 The catalogue and what a source-selected spectrum must supply

## 8.1 What the displayed source assigns

The common finite construction specifies three distinct charged-lepton coefficients, six quark coefficients with color multiplicities, and three zero neutrino coefficients in a 45-component internal kernel. The gauge and Higgs directions belong to the separate conditional bosonic action.

| Channel | Displayed source input | Established finite meaning |
|---|---|---|
| Charged leptons | $Y_e = \sqrt{2}\, g Z_C$ | Three memory-derived singular coefficients on the native charged-lepton slots |
| Up and down flavors | Diagonal powers of 1/6 | Six declared coefficients on native color slots; this displayed choice has identity quark mixing |
| Minimal neutrino slots | Zero | Three zero internal coefficients, without a massless relativistic particle proof |
| Gauge and scalar fields | Canonical one-doublet action, $g_i$, $\lambda_H$, $v$ | Conditional symmetry breaking and quadratic masses; no bosonic defect-state inclusion in the CAR construction |

This is a complete accounting of the displayed model's assignments, not a complete catalogue of nature. Two mathematical distinctions prevent a false physical spectrum. First, the compact four-mode band has a nonzero operator $A = W^\dagger h W$, whose energies enter the exact common operator and cannot be removed to retain the zero-mode mass list. Second, the positive quartic coupling changes the spectral poles. The pixel feedback calculation uses quadratic coefficients, so the compact interacting operator and its exact slice readouts require a recalculated electromagnetic response before a common interacting feedback fixed point can be claimed.

## 8.2 The required attachments

Table 5 states what each missing attachment must deliver. Replacing any of these by measured values calibrates a model; an untuned spectrum identity requires the listed derivation.

These combine into one joint map,

$$\begin{aligned}&\text{(same source, vacuum, clocks, instruments)}\\ &\quad\longrightarrow \text{(total defect energies, current residues, dispersion, vertices)}\\ &\quad\longrightarrow \text{(one response kernel and its pixel fixed point).}\end{aligned}$$

Its first row must supply an autonomous finite-energy defect process in place of an externally fixed texture. Its second must retain all spatial regulator modes until an actual decoupling or continuum theorem controls them. The third must use the first two, including thresholds and physical normalization. These are composition requirements on one permitted design; uniqueness among all possible simulator designs is not a prerequisite.

Table 5: Attachments between the finite stack and a physical spectrum.

| **Attachment** | **Required result** |
|---|---|
| Charged-family source law | Physical scalar Yukawa eigenvalues, phase and flavor frames selected without measured-target fitting |
| Gauge, Higgs and top parameters | Gauge content, running and matching prescription, vacuum scale and the criticality premise derived from the same source |
| Neutral masses and mixing | A nonzero-gap mechanism and complete readout outside the excluded minimal and unshifted-phase branches |
| Physical records | Matched preparation, instruments, clock and estimator, retaining all failure outcomes, with a complete error budget |
| Capacity and refinement | A full-history equivalence or controlled limit for the physical source, beyond the four-address encoding |
| Early-Universe distribution | The relevant state, finite-temperature dynamics, interactions and population observables from the same source |

### 8.3 Interaction, chiral continuum and responsive geometry

Two controls separate spatial propagation from electric motion of a chiral subspace. The complete free three-dimensional overlap Hamiltonian on a flat gauge fiber has $E(p) \to |p|$ and modified axial weight tending to one under refinement; all spatial momenta and internal components are retained. It is vectorlike. A flat fiber has zero Haar measure and supplies no normalizable vacuum for a dynamical gauge field. A separate four-edge compact Abelian flux ring retains all five native charges, with electric energy, matter hopping $J$ and a loop term $\kappa$. For initial flux support $|\ell| \le K_0$, cutoff $K > K_0$, total time $T$ and $r = \lceil (K - K_0 + 1)/\max(1, |q|) \rceil$, a Dyson bound for complete position records, including untouched auxiliary references, gives

$$\mathrm{TV}(\text{complete records}) \le \min\left\{1, 2\sum_{n \ge r} \frac{[(2|J| + |\kappa|)T]^n}{n!}\right\}.$$

At $J = \kappa = 1$, $K = 48$, $K_0 = 0$, $T = 1/2$, this is below 0.000248607 uniformly for the five charges. It controls the stated finite-time preparation and complete observations, not an unbounded-flux ground state without a tail bound.

For the source Euclidean overlap projector $P(\theta)$ on a local admissible charge-six patch, electric differentiation must preserve its range. With $C = [P, P']$, $\nabla = \partial_\theta + C$ obeys $\nabla P = 0$; parallel transport has generator $-C$. For $VV^\dagger = P$, $V^\dagger V = I$, the compressed parent kinetic energy is

$$V^\dagger(-\partial_\theta^2)V = -(\partial_\theta + A)^2 + \Phi, \qquad A = V^\dagger V', \qquad \Phi = V'^\dagger(I - P)V' = V'^\dagger V' + A^2 \succeq 0.$$

Omitting $\Phi$ changes the parent model. A nine-point Dirichlet control keeps all 576 parent components and its 288-dimensional chiral subspace. For positive parent kinetic matrix $K$ and $H_\Lambda = K + \Lambda Q$, let $a_j$ be the eigenvalues of $PKP$. The entire complement is controlled by

$$0 \le a_j - \lambda_j(H_\Lambda) \le \frac{\|PKQ\|^2}{\Lambda - a_k}, \qquad j \le k, \quad \Lambda > a_k.$$

The differential scalar $\Phi$ is positive; the finite-grid difference from a polar-link connection-only discretization need not be. Local frame covariance supplies neither a global anomaly-compatible chiral measure nor physical axial conservation. These spatial, flux and electric constructions are separate controlled models [1].

The quartic result gives a connected interaction on a fixed localized algebra, with a bound uniform in the surrounding radial volume. Its spatial tails and fixed number of localized modes do not define an interacting local $3+1$ field theory. A controlled common field limit must also establish the intended chiral measure, anomaly realization, local gauge constraints, surviving interactions and regulator census. Finite internal anomaly sums are necessary representation checks; a chiral lattice measure has additional locality and integrability requirements [21, 22, 23].

The same matter energy and stress must then enter the geometric response. The companion Einstein implication uses compatible clocks and metric, conserved stress, a geometric modular normalization and controlled fixed-proper-volume generalized-entropy variations [4]. The finite Gibbs first law proved for the interacting localized Hamiltonian is a common-operator identity, and it supplies no geometric area term, small-ball remainder, Ward/Bianchi map or universal Newton normalization.

## Reproduction and mathematical scope

Every statement in this paper has a public executable certificate, proof text or formal module in the project repository [1], including exact characteristic polynomials, rational and integer enclosures, complete line tables, adversarial controls and the source-pinned multiloop replay. They distinguish exact algebra and rational interval proofs from floating spectral checks and empirical comparisons. A source hash identifies a calculation; its mathematical scope is the stated theorem and domain.

## References


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